
StormInsight: Hierarchical Environmental Forcing and Vertical Coupling for Convective Systems Evolution

Jun Chen¹ Yan Fang¹ Minghui Qiu¹ Yueran Qiu¹ Lin Chen¹ Shuxin Zhong¹ Yu Zhang² Kaishun Wu¹

Abstract

Nowcasting forms the first line of defense against rapidly evolving weather hazards, where even minutes of delay can lead to severe societal impacts. However, existing systems predominantly extrapolate 2D radar reflectivity, which struggles under rapid intensification regimes. We introduce *StormInsight*, a multi-scale modeling framework that enables coherent reconstruction of the three-dimensional evolution of convective systems while explicitly conditioning on the ambient environment. *StormInsight* integrates multi-source observations—including radar, satellite, and station—with reanalysis fields through two components: (i) *Storm Evolution Encoder* that explicitly disentangles convective system state from vertical thermodynamic coupling and large-scale environmental forcing; (ii) *Convective System Decoder* that predicts future echos by adaptively aggregating cross-layer interactions conditioned on evolving environmental conditions. To support comprehensive evaluation, we build a new benchmark *StormBench* that integrates observational and reanalysis data across regions. On this benchmark, *StormInsight* consistently achieves the best performance, reducing MAE by 12.4% and improving the mCSI by 34.0%. *Dataset and code:* <https://huggingface.co/datasets/Wow28/StormInsight>.

1. Introduction

Weather nowcasting is the first line of defense against rapidly evolving hazards, underpinning critical decisions in risk mitigation (Nearing et al., 2024) and emergency response (Ravuri et al., 2021; Reichstein et al., 2025). In July

¹the Hong Kong University of Science and Technology (Guangzhou) ²Guangzhou Meteorological Observatory. Correspondence to: Shuxin Zhong <shuxinzhong@hkust-gz.edu.cn>.

2025, Kerrville, Texas experienced a once-in-five-centuries rainfall event, with over 250 millimeters of rain falling within three hours (NEWS, 2025). Although the storm was detected, its abrupt transition to extreme intensity went unrecognized in time; warnings arrived too late, and flash floods claimed over 130 lives. This tragedy highlights a fundamental limitation of current nowcasting: the greatest risk lies not in observing storms, but in missing how they evolve (Zhang et al., 2023).

Existing nowcasting systems formulate the task as radar echo extrapolation, treating weather evolution as a purely visual prediction problem. Representative approaches include ConvLSTMs (Shi et al., 2015), optical flow methods (Bowler et al., 2004), Transformers (Wu et al., 2024a; Gao et al., 2022a), and models trained with Fourier (Lin et al., 2025) or diffusion objectives (Yu et al., 2024). While these methods perform well in dissipation- or translation-dominated scenarios—where flattened radar reflectivity suffices for motion tracking—they consistently struggle under rapid intensification regimes, as such two-dimensional representations lack the information needed to anticipate storm evolution.

Convection evolution is fundamentally a three-dimensional, environment-driven process, rather than a phenomenon observable in the plane alone. The initiation, organization, and dissipation of convection arise from tightly coupled interactions across vertical layers: near-surface moisture transport and wind convergence establish preconditions for storm development (Schumacher & Rasmussen, 2020); mid-level thermodynamic processes regulate updraft strength and latent heat that shape storm structure (Feingold et al., 2010); and upper-layer dynamics govern cloud-top cooling and anvil expansion (Bony et al., 2016). Crucially, these layer-wise interactions are continuously modulated by large-scale environmental forcing—from synoptic-scale ascent to mesoscale convergence—driving the critical transitions between storm genesis, rapid intensification, and dissipation. Any nowcasting approach that fails to account for this vertical coupling and environmental context is therefore inherently limited in its ability to anticipate how convective systems evolve, particularly during high-impact transitions.

However, effectively exploiting multi-altitude observations goes beyond naive layer stacking, as it encoun-

ters two fundamental challenges. **Asynchronous Cross-layer Interactions (C1):** Convective feedbacks propagate across vertical layers on mismatched and state-dependent timescales (Houze Jr, 2004). For instance, near-surface moisture convergence may precede mid-level latent heat release or cloud-top cooling by tens of minutes—or fail to trigger them altogether (Williamson et al., 2019). This temporal asynchrony breaks the locality and alignment assumptions implicit in most multi-layer fusion models, rendering simple stacking or synchronized attention ineffective for capturing causal cross-layer dynamics (Runge et al., 2019). **Environment-Dependent Sensitivity (C2):** The impact of local precursors is not intrinsic, but conditioned on the evolving synoptic-scale environment. Identical low-level convergence patterns can either initiate deep convection or be entirely suppressed under mid-level dry intrusions or unfavorable thermodynamic profiles (Parker et al., 2016; Barton et al., 2025). These context dependence induces a non-stationary input-output relationship, requiring models to dynamically gate local storm evolution by large-scale environmental forcing rather than treating environmental variables as static side information.

To address these challenges, we design *StormInsight*, a meteorology-inspired framework that rethinks nowcasting as environment-conditioned, vertically structured dynamical modeling rather than flat visual extrapolation. To address **C1**, *StormInsight* employs a tripartite encoding scheme (*Storm Evolution Encoder*) that explicitly separates what evolves, how layers interact, and under which atmospheric context this interaction unfolds. Specifically, (i) the *Convective State Encoder* extracts the instantaneous physical states of the system (e.g., hydrometeor growth) from multi-modal observations; (ii) the *Vertical Interaction Encoder* models directional and temporally decoupled cross-layer feedbacks, such as delayed responses between low-level convergence and mid-level latent heating; and (iii) *Atmospheric Environment Encoder* captures synoptic-scale constraints, including wind shear and humidity profiles, that regulate when and where such interactions can manifest. To tackle **C2**, *StormInsight* utilizes a hierarchical modulation strategy (*Convective System Decoder*) that conditions local storm evolution on large-scale atmospheric forcing. A *Global Adapter* enforces synoptic-level constraints on the overall evolution trajectory, while a *Local Adapter* injects vertical feedbacks into frame-level generative inference. Finally, We introduce *StormBench*, a new benchmark that unifies multi-source observations and reanalysis fields to enable rigorous evaluation of environment-aware, three-dimensional convective nowcasting.

Our main contributions are summarized as follows:

- We move beyond 2D radar echo extrapolation and introduce a new formulation that treats convective nowcasting

as a *three-dimensional, environment-aware dynamical reasoning problem*. By integrating multi-source observations with reanalyze fields, we capture the lifecycle of convective systems—including genesis, intensification, and dissipation—rather than merely tracking surface-level motion.

- **StormInsight** designs two components: (i) the *Storm Evolution Encoder* disentangles convective dynamics into system state, vertical thermodynamic exchanges, and synoptic-scale environmental context, enabling asynchronous and directional cross-layer reasoning; (ii) the *Convective System Decoder* hierarchically modulates generative inference with these representations, allowing identical local precursors to yield different outcomes under varying environmental forcings.

- We introduce *StormBench*, a new large-scale benchmark that unifies satellite, radar, station observations, and atmospheric reanalyze data across multiple regions to support rigorous evaluation of three-dimensional, environment-conditioned nowcasting. Extensive experiments show that *StormInsight* consistently outperforms all baselines, reducing MAE by 12.4%, and improving mCSI by 34.0%, while also delivering robust gains on downstream tasks, such as rainfall field reconstruction.

2. Related Work

2.1. Vertical Structure of Convection System

Convective systems exhibit vertical stratification (Emanuel, 1994). The near-surface layer is driven by moisture and wind convergence (Schumacher & Rasmussen, 2020), which can be captured by weather stations; the mid-levels are characterized by updrafts and latent heat release, with radar capturing convective organization (Schumacher et al., 2004; Feingold et al., 2010); and the upper layer is marked by ice-phase sedimentation and anvil expansion, identified via satellite (Choi et al., 2010; Bony et al., 2016). Each layer possesses distinct physical mechanisms and interacts vertically to form a system: upper-level divergence can enhance mid-level ascent (Squitiery et al., 2023), while localized surface lifting can trigger new convective cells (Davolio et al., 2018), collectively governing the initiation, maintenance, and decay of convection (Houze Jr, 2004). Despite decades of meteorological research, most current ML-based methods still fail to incorporate vertical structure.

2.2. Atmospheric Environment of Convection System

The evolution of convective systems is modulated by the surrounding atmospheric environment, which can be captured by reanalyzed data and governs whether convection can be initiated, intensified, or dissipated (Williamson et al., 2019). Thermodynamically, warm, moist conditions favor persistent updrafts, whereas mid-level dry intrusions

can rapidly weaken convection (Parker et al., 2016; Berg et al., 2013). Dynamically, mesoscale convergence controls the vertical motion necessary to release instability, shaping whether convection dissipates or organizes into longer-lived systems (Barton et al., 2025). At larger scales, synoptic forcing and moisture transport modulate ascent and moisture supply, thereby controlling the overall life cycle and spatial extent of convective systems (Dacre et al., 2015; Baruah et al., 2025). In this way, the atmospheric environment defines both the boundary conditions and the energetics through which convection evolves from initiation to decay.

2.3. Video-Centric Echo Extrapolation

Radar echo extrapolation is conventionally framed as a video prediction task. Early recurrent models (e.g., ConvLSTM (Shi et al., 2015), PredRNN-V2 (Wang et al., 2022)) introduce spatiotemporal memory for long-range dependencies, but struggle with parallelism and produce blurry outcomes in high-resolution convection. To overcome these limitations, recurrent-free approaches have gained prominence. SimVP (Gao et al., 2022b) decouples spatial and temporal modeling through a pure convolutional framework; transformer-based models such as Earthformer (Gao et al., 2022a) and EarthFarSeer (Wu et al., 2024a) capture long-range context via self-attention; AlphaPre (Lin et al., 2025) disentangles amplitude and phase to enhance spatial structure; and generative approaches like Prediff (Gao et al., 2023) model echo evolution as a diffusion-based generative process. Most methods treat nowcasting as a 2D texture deformation problem and overlook the influence of the large-scale environment. *This flattening erases the storm’s vertical structure, causing distinct phenomena such as intensifying updrafts and collapsing cores to appear identical.*

3. Methodology

3.1. Problem Formulation

We reformulate radar echo extrapolation as a meteorological-informed forecasting problem, in which the model is required to simulate interactions across vertically stratified layers while accounting for the influence of the atmospheric environment. Given the input sequence $\mathcal{X} = \{\mathbf{X}_H, \mathbf{X}_M, \mathbf{X}_L, \mathbf{X}_{\text{env}}\}$, where $\mathbf{X}_H$, $\mathbf{X}_M$, and $\mathbf{X}_L$ correspond to satellite, radar, and station that describe meteorological processes at different layers, and $\mathbf{X}_{\text{env}}$ denotes the reanalysis data introducing background environment information. Our objective is to learn a function $\mathbf{f}_\theta$ that predicts the future reflectivity field by modeling cross-layer coupling under varying ambient conditions. Formally, we seek:

$$\hat{\mathcal{Y}} = \mathbf{f}_\theta(\mathcal{X}), \quad \hat{\mathcal{Y}} = \{\mathbf{Y}_{T+1}, \dots, \mathbf{Y}_{T+K}\}, \quad (1)$$

where K denotes the forecast horizon.

3.2. Overall Architecture

To model convection as a vertically stratified, environmental-driven dynamical process, we introduce *StormInsight*, a meteorology-inspired framework composed of two tightly coupled components (see Figure 1):

- **Storm Evolution Encoder** constructs a structured representation of the convective system by disentangling (i) within-layer physical states, (ii) directional cross-layer interactions, and (iii) synoptic-scale environmental context from heterogeneous observations and reanalysis data.
- **Convective System Decoder** performs environment-aware temporal propagation conditioned on the encoded environmental and cross-layer structure, adaptively fusing within- and cross-layer signals to infer future reflectivity fields.

3.3. Storm Evolution Encoder

The *Storm Evolution Encoder* represents convective evolution through a structured decomposition along three complementary dimensions: (i) *Convective State Encoder* captures within-layer storm states and morphology from multi-modal observations; (ii) *Vertical Interaction Encoder* models directional thermodynamic coupling across layers, including moisture convergence, latent heat release, and ice-phase processes; (iii) *Atmospheric Environment Encoder* encodes the mesoscale background conditions—including moisture stratification and geopotential structure—that regulate when and how these interactions manifest.

3.3.1. CONVECTIVE STATE ENCODER

Universal Gridded Projection. To reconcile the heterogeneous sampling densities of observation sources, we project all inputs onto a shared latitude-longitude grid via a unified gridded representation. Given raw inputs $x_n = \{x_H, x_M, x_L\}$ from high-, mid-, and low-altitude modalities, we define modality-specific projections ψ_n that map each input into spatially aligned tensor:

$$\{\mathcal{G}_H, \mathcal{G}_M, \mathcal{G}_L\} = \{\psi_n(x_n)\}_{n \in \{H, M, L\}}. \quad (2)$$

Each $\mathcal{G}_n \in \mathbb{R}^{C_n \times V \times W}$ is defined on a common $V \times W$ grid, enabling consistent cross-modal and cross-layer reasoning.

We implement ψ_n using a SetConv-based projection layer (Allen et al., 2025), which supports both regular (satellite, radar) and irregular (station) spatial sampling. Specifically,

$$\mathcal{G}_n = \psi_n(x_n) = \text{SetConv}(\mathbf{h}_{loc}, \mathbf{h}_n) \quad (3)$$

where $\mathbf{h}_{loc}$ denotes spatial coordinates and $\mathbf{h}_n$ represents modality-specific physical attributes. A learnable RBF kernel enables adaptive interpolation, allowing sparse and dense observations to be mapped onto a unified spatial resolution in a data-driven manner.

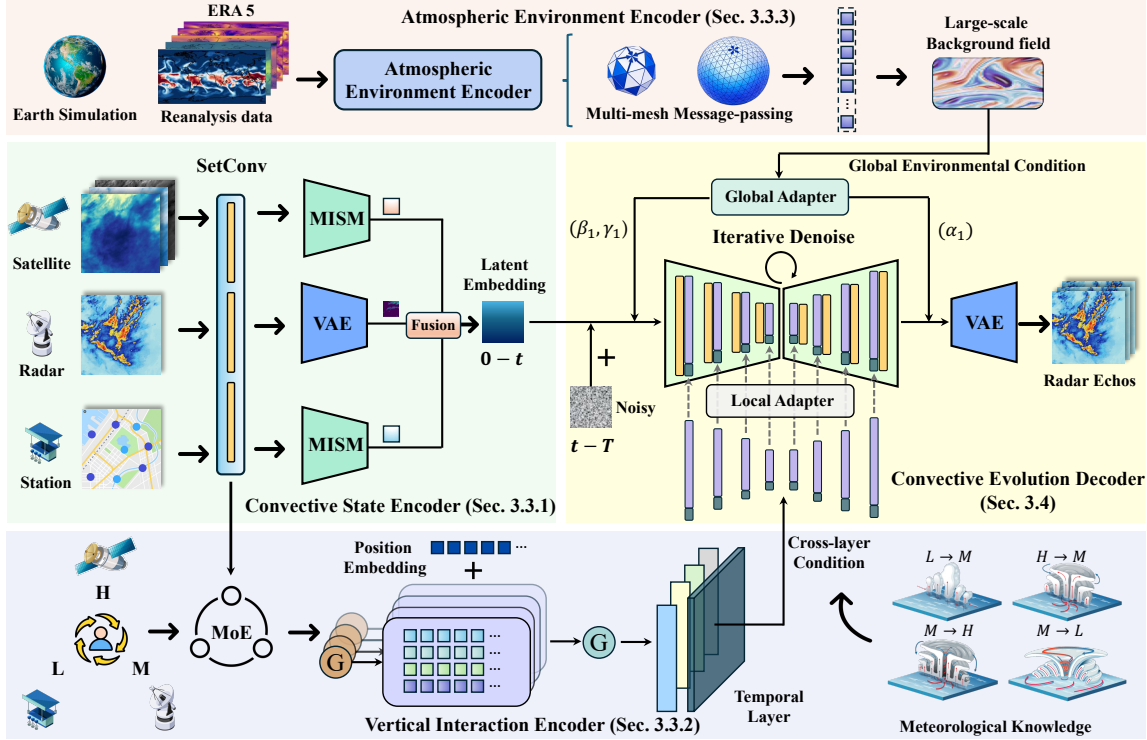


Figure 1. The Framework of StormInsight.

Multi-scale Feature Extraction. Convective systems exhibit both large-scale stratiform organization and localized convective cores. To capture this multi-scale structure, we apply a Multi-Scale Inception Module (MSIM) $\phi_{\text{MSIM}}(\cdot)$ (Szegedy et al., 2016) to each satellite and station slice $\mathbf{X}_t^i \in \mathbb{R}^{1 \times V \times W}$ from the i -th attribution. MSIM uses strided convolutions with varying kernel sizes to extract feature at multiple spatial resolutions, while an average-pooling branch preserves low-frequency background signals. The output of $\phi_{\text{MSIM}}(\mathbf{X}_t^i) \in \mathbb{R}^{C_{\text{out}}}$, $i \in \{H, L\}$ encodes the patch-level representation for spatial region p :

$$\mathbf{E}_t^{(i,p)} = \sum_{k \in \{3, \dots, n\}} \text{Conv}_{k \times k}(\text{Conv}_{1 \times 1}(\mathbf{X}_t^i)) + \text{AvgPool}(\mathbf{X}_t^i). \quad (4)$$

The radar modality $\mathcal{G}_{\text{rad}}$ is uniquely mapped into a latent space via a VAE encoder $\mathcal{E}_\theta$ (Kingma & Welling, 2013):

$$\mathbf{z}_M = \mathcal{E}_\theta(\mathcal{G}_M). \quad (5)$$

To form a system-level representation, we fuse the variational radar embedding with the deterministic representations from MSIM. The representation $\mathbf{Z}_t$ for spatial region p is constructed by a FiLM module (Perez et al., 2017) in the latent space:

$$\mathbf{Z}_{\text{sys}}^{(p)} = \text{FiLM}(\mathbf{z}_M^{(p)}, \mathbf{E}_t^{(i,p)}) = \gamma_t^{(p)} \cdot \mathbf{z}_M^{(p)} + \beta_t^{(p)} \quad (6)$$

This fused embedding $\mathbf{Z}_{\text{sys}}$ is used to reconstruct the target radar field, ensuring the generative process is constrained by the physical context provided by satellite and station data.

3.3.2. VERTICAL INTERACTION ENCODER

Inspired by mesoscale meteorology (Bony et al., 2015), *Vertical Interaction Encoder* models cross-layer pathways observed in convective development (details in Appendix B):

- $\mathcal{E}_{M \rightarrow H}$: Mid-level moisture convergence and latent heat release enhance updraft buoyancy, accelerating the vertical transport of air masses to the upper troposphere.
- $\mathcal{E}_{H \rightarrow M}$: Top-down feedback, such as anvil spreading and radiative cooling at cloud tops, can modulate mid-level upward motion and trigger secondary convection.
- $\mathcal{E}_{M \rightarrow L}$: Mid-level convective downdrafts and inflows disrupt boundary-layer thermodynamics and reshape near-surface convergence structures.
- $\mathcal{E}_{L \rightarrow M}$: Surface-driven initiation where boundary-layer heating and moistening weaken convective inhibition, triggering new convective cells.

To encode these directional pathways, we design an expert-driven Mixture-of-Experts (MoE) component. The process begins by encoding each layer using MSIM: $\mathbf{F}_t^i = \phi_{\text{MSIM}}(\mathbf{X}_t^i)$, $i \in \{H, M, L\}$. Each expert $\mathcal{E}_{i \rightarrow j}(\cdot)$, built upon a Transformer backbone, specializes in learning how

signals from source layer i modulate the future evolution of target layer j :

$$\mathbf{Z}_{T:T+K}^{i \rightarrow j} = \mathcal{E}_{i \rightarrow j}(\mathbf{F}_{1:T}^i), \quad i \neq j \quad (7)$$

To adaptively integrate forecasts from different time windows, we design the Gate-of-Time module. This module addresses a challenge in weather forecasting—asynchronous influence between cross-layer variables:

$$\begin{aligned} \mathbf{E}_t^{i \rightarrow j} &= \sigma(\mathbf{W}_g[\mathbf{Z}_{T:T+K}^{i \rightarrow j}; \mathbf{F}_{1:T}^j] + \mathbf{b}_g), \\ \alpha_t^{(j)} &= \text{MLP}(\mathbf{F}_{1:T}^j) \end{aligned} \quad (8)$$

where $\mathbf{W}_g$ and $\mathbf{b}_g$ are learnable parameters, and $\sigma(\cdot)$ is the sigmoid function. The final prediction of each expert is decoded by a lightweight decoder(Gao et al., 2022b):

$$\hat{\mathbf{X}}_t^j = \alpha_t^{(j)} \cdot \text{Decoder}(\mathbf{E}_t^{i \rightarrow j}) \quad (9)$$

Each expert is trained with a directional reconstruction loss:

$$\mathcal{L}_{i \rightarrow j} = \frac{1}{K} \sum_{t=T}^{T+K} \left\| \hat{\mathbf{X}}_t^j - \mathbf{X}_t^j \right\|_2^2, t \in [T, T+K] \quad (10)$$

3.3.3. ATMOSPHERIC ENVIRONMENT ENCODER

To incorporate multi-level thermodynamic context from reanalysis data, we encode environmental information based on Multi-mesh Message Passing (Lam et al., 2023). Given the atmospheric field $\mathbf{X}_{env} \in \mathbb{R}^{N \times C \times H \times W}$ with N pressure levels and C variables, the encoder first generates node-wise embeddings through a local feature projection:

$$h_v^{(0)} = \text{MLP}(\mathbf{X}_{env}[v]), \quad h_v^{(0)} \in \mathbb{R}^{D_0}, \quad (11)$$

where $v \in V_{\text{mesh}}$ indexes the nodes on a multi-resolution spherical mesh. To capture vertical thermodynamic and horizontal synoptic dependencies from reanalysis data:

$$m_{u \rightarrow v}^{(k)} = \psi(h_u^{(k)}, h_v^{(k)}, e_{uv}), \quad (12)$$

$$m_v^{(k)} = \sum_{u \in \mathcal{N}(v)} m_{u \rightarrow v}^{(k)}, \quad (13)$$

$$h_v^{(k+1)} = \eta(h_v^{(k)}, m_v^{(k)}), \quad (14)$$

where e_{uv} encodes multi-resolution geometric relations and cross-level adjacency, $\mathcal{N}(v)$ denotes the neighbors of node v , and η includes residual updates and normalization. After K layers, the encoder outputs an environment embedding:

$$\mathbf{Z}_{env} = \{h_v^{(K)} \mid v \in V_{\text{mesh}}\}, \quad \mathbf{Z}_{env} \in \mathbb{R}^{D \times |V_{\text{mesh}}|}. \quad (15)$$

This encoding retains high-order vertical dependencies and synoptic-scale dynamical structure, providing a physically grounded environmental context for forecasting.

3.4. Convective System Decoder

Convective System Decoder performs environment-conditioned temporal propagation by adaptively fusing system state, directional cross-layer feedbacks (e.g., radiative cooling, vertical coupling), and large-scale atmospheric context to model convective initiation, evolution, and decay.

3.4.1. HIERARCHICAL PHYSICAL CONDITIONING

Global Adapter. Large-scale thermodynamic and kinematic forcings extracted from reanalysis are injected as global conditioning via AdaLN (Peebles & Xie, 2022) at both the beginning and the end of the flow matching:

$$\text{AdaLN}(\mathbf{h}, \mathbf{Z}_{\text{env}}) = \gamma(\mathbf{Z}_{\text{env}}) \odot \text{Norm}(\mathbf{h}) + \beta(\mathbf{Z}_{\text{env}}), \quad (16)$$

where γ and β are learnable scale and shift parameters derived from $\mathbf{Z}_{\text{env}}$, $\mathbf{h}$ is the latent representation of backbone. This enables the model to infer convective evolution under environmental constraints.

Local Adapter. To condition the expert fusion on layer-specific convective states, we introduce a gate mechanism:

$$\mathbf{Z}'_{\text{expert}} = \mathcal{C}_t \odot \sigma(\mathbf{W}_g \mathcal{C}_t + \mathbf{b}_g). \quad (17)$$

where the cross-layer knowledge is adaptively injected into the inference of the intra-layer changes. $\mathcal{C}_t = [\mathbf{C}_t^{i \rightarrow j}]_{i \neq j, i, j \in \{H, M, L\}}$ denote the concatenated tensor of vertical interaction features at timestep t . Sinusoidal position encodings are added to $\mathbf{Z}'_{\text{expert}}$ to preserve temporal order.

To decompose these cross-layer features into frame-level conditions, we employ a temporal Transformer (Guo et al., 2024). The aggregated features $\mathbf{Z}_{\text{expert}}$ are processed through self-attention blocks:

$$\mathbf{Z}_{\text{expert}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}, \quad (18)$$

where $\mathbf{Q} = \mathbf{W}^Q \mathbf{Z}'_{\text{expert}}$, $\mathbf{K} = \mathbf{W}^K \mathbf{Z}'_{\text{expert}}$, and $\mathbf{V} = \mathbf{W}^V \mathbf{Z}'_{\text{expert}}$ are projections, and d_k is the dimension of the key vectors. This attention mechanism aggregates temporal context across frames to refine cross-layer representations, which are then injected into every attention block of the flow matching backbone via AdaLN, enabling frame-wise reasoning over layer interactions.

3.4.2. CONDITIONAL FLOW MATCHING

Let $\mathbf{Z}_{\text{cond}} = \{\mathbf{Z}_{\text{sys}}, \mathbf{Z}_{\text{env}}, \mathbf{Z}_{\text{expert}}\}$ represent the aggregate meteorological conditions. We adopt the Conditional Flow Matching framework (Lipman et al., 2022) to learn a deterministic transport map from a standard Gaussian prior $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to the target latent distribution $\mathbf{z}_1$.

The generative process is governed by the ODE:

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t, \mathbf{Z}_{\text{cond}}), \quad t \in [0, 1], \quad (19)$$

where $\mathbf{v}_\theta$ is a neural vector field. The final radar prediction is obtained by integrating Eq. (19) from $t = 0$ to 1 via an ODE solver and decoding the result using the frozen VAE decoder: $\hat{\mathcal{Y}} = \mathcal{D}(\mathbf{z}_1)$.

Training Objective. We construct the conditional probability path using the *optimal transport* interpolant $\mathbf{z}_t = (1 - t)\mathbf{z}_0 + t\mathbf{z}_1$. This path corresponds to trajectories with a constant target velocity $\mathbf{u}_t(\mathbf{z}_t | \mathbf{z}_1, \mathbf{z}_0) = \mathbf{z}_1 - \mathbf{z}_0$. Consequently, the model is trained to regress this vector field:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, \mathbf{z}_0, \mathbf{z}_1} \left[\|\mathbf{v}_\theta(\mathbf{z}_t, t, \mathbf{Z}_{\text{cond}}) - (\mathbf{z}_1 - \mathbf{z}_0)\|_2^2 \right], \quad (20)$$

where $t \sim \mathcal{U}[0, 1]$, $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\mathbf{z}_1$ represents the latent representation of ground-truth data sample.

3.4.3. METEOROLOGICAL INTERPRETATION

To elucidate how meteorological constraints shape the generation trajectory, we decompose the conditioning mechanism into global forcing and local guidance.

Global Environment Forcing. The velocity field decomposes into intrinsic dynamics and an environmental bias:

$$\mathbf{v}_\theta(\mathbf{z}_t, t, \mathbf{Z}_{\text{cond}}) = \mathbf{v}_{\text{core}}(\mathbf{z}_t, t, \mathbf{Z}_{\text{sys}}, \mathbf{Z}_{\text{expert}}) + \underbrace{\mathcal{F}_\theta(\mathbf{Z}_{\text{env}})}_{\text{Global Drift}}. \quad (21)$$

Isomorphic to the atmospheric equation $\frac{\partial \mathcal{S}}{\partial t} = \mathcal{D}(\mathcal{S}) + \mathcal{F}_{\text{env}}$ (Kalnay, 2003), the term $\mathcal{F}_\theta$ acts as a global external forcing that shifts the entire transport trajectory.

Vertical Interaction. Unlike the global drift, the cross-layer features $\mathbf{Z}_{\text{expert}}$ are injected into every attention layer of the core network $\mathbf{v}_{\text{core}}$, providing fine-grained constraints:

$$\mathbf{v}_{\text{core}}(\mathbf{z}_t, \dots) = \mathbf{v}_{\text{core}} \left(\mathbf{z}_t \cdot \underbrace{\gamma(\mathbf{Z}_{\text{expert}})}_{\text{Scale}} + \underbrace{\beta(\mathbf{Z}_{\text{expert}})}_{\text{Shift}} \right). \quad (22)$$

$\mathbf{Z}_{\text{expert}}$ modulates the geometry of the vector field, ensuring that the evolution is constrained by the vertical structure.

4. Evaluation

This section presents a comprehensive evaluation of StormInsight by addressing the following questions:

- **RQ1:** What is StormInsight’s overall performance?
- **RQ2:** How does each component affect performance?
- **RQ3:** How robust is StormInsight across lead times?
- **RQ4:** How does StormInsight perform during storm initiation and dissipation phases?

4.1. Evaluation Settings.

4.1.1. DATASETS

We evaluated StormInsight on our self-constructed dataset StormBench (Appendix A), which containing multi-source observations (satellites, radars, ground weather stations) and reanalysis data, covering weather events in the United States (2017-2019) and France (2016-2018). We downscale the spatial size to 128×128 , due to the limitation of our computation resource. Please refer to the Appendix D.1 for information on variables selection and detailed configuration. StormInsight predicts the long-term precipitation evolution for 25 frames based on 13 frames.

4.1.2. BASELINES.

We compare StormInsight against ten competitive baselines, including **ConvLSTM** (Shi et al., 2015), **PredRNN-V2** (Wang et al., 2022), **PhyDNet** (Guen & Thome, 2020), **SimVP** (Gao et al., 2022b), **Earthformer** (Gao et al., 2022a), **NowcastNet** (Zhang et al., 2023), **PastNet** (Wu et al., 2024b), **Earthfarseer** (Wu et al., 2024a), and **AlphaPre** (Lin et al., 2025) (detailed in Appendix D.2).

4.1.3. METRICS.

To evaluate both the quantitative accuracy and physical fidelity of the predicted radar fields, we adopt a suite of complementary metrics: **MAE** for distributional bias, **CSI** and **HSS** for event-based storm detection, and **SSIM** and **PSNR** for spatial-perceptual consistency. The computation of each metric is detailed in the Appendix D.3.

4.1.4. IMPLEMENTATION DETAILS.

The VAE component is trained following the procedure described in (Gao et al., 2023). StormInsight is trained on four NVIDIA H100 GPUs and eight Intel(R) Xeon(R) Platinum 8358P CPUs (2.60 GHz) for 500 epochs with a batch size of 64. For the ODE solver, we use the Euler method with 20 discretization steps, chosen for its simplicity and computational efficiency.

4.2. Overall Performance (RQ1)

Table 1 presents the comprehensive evaluation and reveals three insights:

- **Non-recurrent models yield better storm detection.** Non-recurrent architectures (e.g., SimVP and Earthformer) achieve the highest mCSI and mHSS scores in the USA experiments. Their combination of global attention and spatial filtering modules is effective for capturing storm structures. They remain strong under both broad, low-intensity echoes (CSI₁₉) and sparse severe convection (CSI₂₁₉), indicating that multi-scale feature benefits the modeling of

Table 1. Performance comparison across different regions. Best results are in **bold**.

Model	USA								France							
	MAE↓	mCSI↑	mHSS↑	CSI ₁₆ ↑	CSI ₇₄ ↑	CSI ₂₁₉ ↑	PSNR↑	SSIM↑	MAE↓	mCSI↑	mHSS↑	CSI ₁₉ ↑	CSI ₂₈ ↑	CSI ₄₇ ↑	PSNR↑	SSIM↑
ConvLSTM	15.626	0.282	0.361	0.585	0.500	0.142	22.000	0.419	1.822	0.038	0.064	0.163	0.028	0.000	40.231	0.840
PredRNN-V2	15.685	0.360	0.466	0.625	0.532	0.285	23.019	0.461	1.860	0.052	0.086	0.195	0.059	0.000	40.167	0.833
PhyDNet	16.664	0.324	0.410	0.606	0.521	0.214	22.779	0.430	2.199	0.013	0.023	0.058	0.005	0.000	39.069	0.790
SimVP	14.176	0.403	0.515	0.657	0.566	0.340	23.888	0.530	1.600	0.106	0.169	0.293	0.174	0.001	40.164	0.861
Earthformer	14.935	0.369	0.469	0.648	0.544	0.298	23.397	0.489	1.513	0.087	0.138	0.271	0.130	0.000	40.376	0.865
NowcastNet	18.964	0.318	0.404	0.624	0.513	0.108	21.361	0.503	1.957	0.036	0.061	0.109	0.046	0.001	37.202	0.819
PastNet	15.330	0.377	0.482	0.629	0.569	0.265	23.232	0.486	1.609	0.104	0.169	0.273	0.155	0.005	39.959	0.852
Earthfarseer	14.845	0.366	0.470	0.640	0.551	0.236	23.352	0.511	1.693	0.080	0.130	0.247	0.113	0.002	39.694	0.844
AlphaPre	15.923	0.361	0.451	0.626	0.544	0.244	23.422	0.505	1.549	0.094	0.151	0.255	0.144	0.002	40.004	0.861
Ours	12.608	0.540	0.632	0.673	0.552	0.361	23.651	0.573	0.980	0.258	0.380	0.448	0.354	0.087	30.089	0.811

 Table 2. Ablation study on the key components of our method. The best results among variants are highlighted in **bold**.

Variant	USA								France							
	MAE↓	mCSI↑	mHSS↑	CSI ₁₆ ↑	CSI ₇₄ ↑	CSI ₂₁₉ ↑	PSNR↑	SSIM↑	MAE↓	mCSI↑	mHSS↑	CSI ₁₉ ↑	CSI ₂₈ ↑	CSI ₄₇ ↑	PSNR↑	SSIM↑
w/o AEE	21.043	0.281	0.319	0.411	0.343	0.172	18.731	0.273	1.939	0.119	0.173	0.253	0.198	0.033	21.435	0.378
w/o VIE	15.162	0.410	0.497	0.559	0.429	0.211	20.930	0.498	1.397	0.174	0.270	0.344	0.248	0.040	25.126	0.689
w Radar	21.980	0.216	0.233	0.355	0.230	0.112	17.199	0.250	2.025	0.091	0.126	0.218	0.133	0.021	23.868	0.346
w/o $\mathcal{E}_{H \rightarrow M}$	12.960	0.494	0.590	0.636	0.510	0.321	23.337	0.573	1.194	0.209	0.320	0.392	0.295	0.061	29.166	0.792
w/o $\mathcal{E}_{L \rightarrow M}$	12.970	0.495	0.592	0.637	0.511	0.318	23.321	0.572	1.195	0.210	0.321	0.393	0.296	0.060	29.139	0.790
w/o $\mathcal{E}_{M \rightarrow H}$	12.799	0.530	0.622	0.660	0.543	0.342	23.423	0.562	1.179	0.225	0.337	0.407	0.314	0.065	29.310	0.777
Ours	12.608	0.540	0.632	0.673	0.552	0.361	23.651	0.573	0.980	0.258	0.380	0.448	0.354	0.087	30.089	0.811

spare strong storm.

• **Recurrent architectures exhibit generally weaker performance.** Multi-source inputs substantially degrade their accuracy (e.g., ConvLSTM), suggesting limited capacity for cross-modal fusion. Meanwhile, Transformer- and CNN-based models such as Earthformer and AlphaPre continue to improve, reflecting their stronger spatial alignment and channel-wise decoupling capabilities.

• **StormInsight consistently outperforms all baselines.** StormInsight yields the best performance across regions and metrics, reducing MAE by 12.4% and improving mCSI by up to 34.0%. These results highlight *the importance of explicitly modeling cross-layer convective interactions and large-scale environmental fields*.

4.3. Ablation Study (RQ2)

To evaluate each design, we conduct an ablation study:

- w/o AEE: Remove *Atmospheric Environment Encoder*, disabling large scale environment influence.
- w/o VIE: Remove *Vertical Interaction Encoder* to assess stratified processing.
- w Radar: Remove AEE, VIE and other modality input within MISIM to evaluate the performance only with radar.
- w/o $\mathcal{E}_{i \rightarrow j}$: For each fine-grained ablation, one expert (e.g., $\mathcal{E}_{H \rightarrow M}$, $\mathcal{E}_{L \rightarrow M}$, $\mathcal{E}_{M \rightarrow H}$, $\mathcal{E}_{M \rightarrow L}$) is removed to assess diverse influence of different vertical interactions.

The results yield several key observations (Table 2):

- **Environmental forcing regulates convection.**

StormInsight w/o AEE yields a noticeable decrease in CSI₂₁₉, indicating that the large-scale thermodynamic and dynamic environment regulates the convective life cycle, and its influence becomes dominant during the strengthening phase.

• **Explicit vertical interaction modeling is essential.** StormInsight w/o VIE results in consistent decreases across all metrics, with the largest degradation appearing in mCSI. This underscores the necessity of explicit cross-layer interaction modeling for capturing weak precursors (e.g., low-level convergence) during convective initiation.

• **Multi-modal Synergy Stabilizes forecasting.** w Radar produces the largest degradation. Radar reflectivity cannot provide all the physical context required for storm evolution, whereas observational modalities and reanalysis fields offer complementary information that stabilizes long lead-time prediction.

• **Expert fusion ensures robust vertical interaction.** Removing any expert (e.g., w/o $\mathcal{E}_{H \rightarrow M}$) causes degradation of performance. This supports the hypothesis that convective development emerges from coordinated cross-layer interactions rather than from a single dominant pathway.

4.4. Forecast Horizon Analysis (RQ3)

Figure 2 illustrates model performance across forecast horizons from 20 to 120 minutes, using both pix-level error (MAE) and structure-aware accuracy (SSIM). As expected, performance degrades as lead time increases, but the rate of decline varies: recurrent-based models (e.g., ConvLSTM, PredRNN-V2) deteriorate faster than recurrent-free ones, reflecting weaker spatial generalization over longer

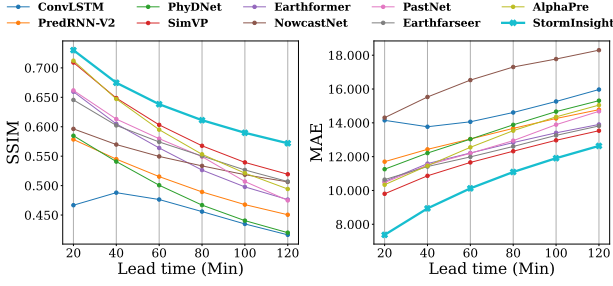


Figure 2. SSIM/MAE Trends across Lead Times

horizons. StormInsight shows a slower decay, which demonstrates its capability to temporal degradation: sensing convective system well after conventional echo-based models fail.

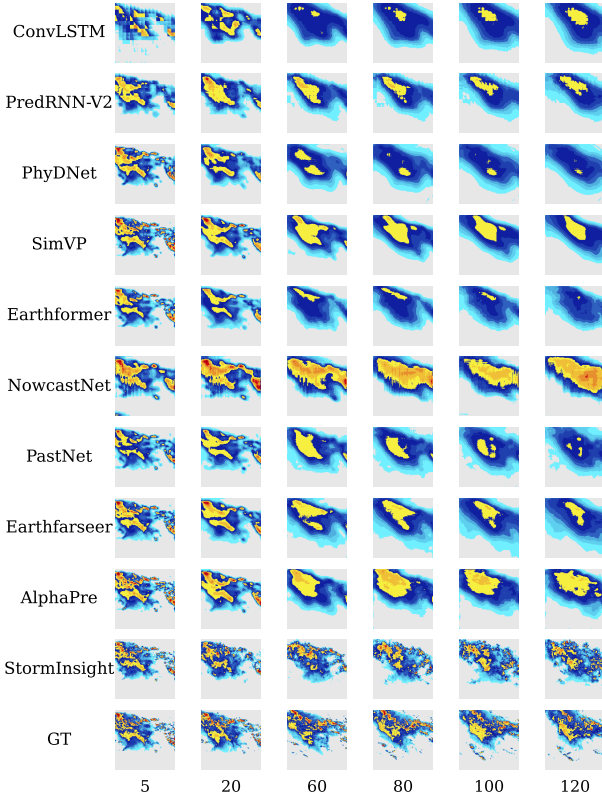


Figure 3. The Convective Development Case.

4.5. Case Study: Capturing Convective Development and Dissipation (RQ4)

To examine the physical validity of StormInsight during storm evolution, we conduct two state studies representing distinct lifecycle phases: convective *intensification* and *dissipation*, selected based on the Convective Area Variation Rate (CAVR) (Pan et al., 2021), a physically grounded indicator of echo dynamics. Figure 3 shows intensifying cases, where StormInsight consistently produces more coherent storm structures than state-of-the-art baselines (More

visualization examples, including dissipating and intensifying storms are provided in Appendix E.4). In intensifying phase (positive CAVR), StormInsight captures the emergence and localization of high-intensity convective cores that competing models often diffuse or miss. In dissipating phase (negative CAVR), it preserves fine-grained decay patterns while avoiding spatial drift. These cases highlight StormInsight’s ability to track the full convective lifecycle with high spatial and semantic fidelity—addressing a long-standing challenge in operational nowcasting.

5. Limitation and Future Work

While StormInsight performs well across tasks, yet leaves room for improvement: (i) Our current framework primarily operates on multi-layer 2D representations, without fully exploiting the volumetric 3D structure of radar echoes. Integrating full 3D scans would allow for a more granular characterization of the storm core’s vertical development and updraft dynamics. (ii) The potential of polarimetric radar variables (e.g., differential reflectivity Z_{DR} and specific differential phase K_{DP}) remains untapped.

Despite these limitations, the multi-source framework and dataset, integrating satellite, radar, station, and reanalysis data, will help advance more physically grounded and interpretable models for severe-weather nowcasting.

6. Conclusion

In this study, we introduce StormInsight, a meteorology-inspired framework that reframes convective nowcasting as a 3D spatiotemporal, environment-conditioned spatiotemporal modeling problem. By explicitly modeling asynchronous cross-layer interactions and their dependence on large-scale environment forcing, StormInsight departs from conventional reflectivity-driven extrapolation and captures dynamical mechanisms underlying storm evolution. StormInsight consists of (i) a Storm Evolution Encoder that disentangles physical states, vertical thermodynamic coupling, and environmental context, and (ii) a Convective System Decoder that performs environment-aware temporal propagation to predict future reflectivity fields. To support rigorous evaluation, we construct StormBench, a new benchmark integrating multi-source observations and reanalysis data across different regions. Extensive experiments on StormBench demonstrates that StormInsight consistently outperforms state-of-the-art baselines, improving mCSI by 34.0%. We hope this work encourages the community to move beyond two-dimensional echo extrapolation toward physically grounded, interpretable, and environment-aware models for severe weather nowcasting.

Impact Statement

This work integrates multi-source observational and reanalysis datasets to improve data-driven weather nowcasting. Accurate and timely short-term weather forecasts can benefit a wide range of societal applications, including disaster preparedness, emergency response, and aviation safety. The models developed in this study are designed to advance scientific understanding and operational forecasting capabilities rather than to support high-stakes decision-making autonomously. As such, we do not foresee direct ethical risks associated with model deployment, and the potential broader societal impact is expected to be positive, particularly in the context of climate resilience and extreme weather mitigation. This paper presents work whose goal is to advance the field of machine learning for environmental science, with no foreseeable negative societal consequences beyond those already well understood in scientific data modeling.

Acknowledgments

This work is supported partly by Guangdong Provincial Key Lab of Integrated Communication, Sensing and Computation for Ubiquitous Internet of Things (No.2023B121010007), China NSFC Grant (No.62472366), the Project of DEGP (No.2024GCZX003, 2023KCXTD042), 111 Center (No.D25008), Shenzhen Science and Technology Foundation (ZDSYS20190902092853047), Guangdong-Hong Kong-Macao Greater Bay Area Academy of Meteorological Research (GHMA2025Y03). Guangdong Basic and Applied Basic Research Foundation (2025A1515510015). Guangzhou Smart Meteorology Technology Collaborative Innovation Team - Low-Altitude Flight Meteorological Service Support Technology Innovation Team (NO.202502).

References

- Allen, A., Markou, S., Tebbutt, W., Requeima, J., Bruinsma, W. P., Andersson, T. R., Herzog, M., Lane, N. D., Chantry, M., Hosking, J. S., and Turner, R. E. End-to-end data-driven weather prediction. *Nature*, 641:1172 – 1179, 2025. URL <https://api.semanticscholar.org/CorpusID:277190751>.
- Barton, E. J., Klein, C., Taylor, C. M., Marsham, J., Parker, D. J., Maybee, B., Feng, Z., and Leung, L. R. Soil moisture gradients strengthen mesoscale convective systems by increasing wind shear. *Nature Geoscience*, pp. 1–7, 2025.
- Baruah, P. P., Neena, J., and Suhas, E. The role of tropical synoptic-scale disturbances in modulating the strength of east pacific hadley circulation. *Journal of Geophysical Research: Atmospheres*, 130(9):e2024JD042980, 2025.
- Berg, P., Moseley, C., and Haerter, J. O. Strong increase in convective precipitation in response to higher temperatures. *Nature Geoscience*, 6(3):181–185, 2013.
- Bony, S., Stevens, B., Frierson, D. M., Jakob, C., Kageyama, M., Pincus, R., Shepherd, T. G., Sherwood, S. C., Siebesma, A. P., Sobel, A. H., et al. Clouds, circulation and climate sensitivity. *Nature Geoscience*, 8(4): 261–268, 2015.
- Bony, S., Stevens, B., Coppin, D., Becker, T., Reed, K. A., Voigt, A., and Medeiros, B. Thermodynamic control of anvil cloud amount. *Proceedings of the National Academy of Sciences*, 113(32):8927–8932, 2016.
- Bowler, N. E., Pierce, C. E., and Seed, A. Development of a precipitation nowcasting algorithm based upon optical flow techniques. *Journal of Hydrology*, 288(1-2):74–91, 2004.
- Choi, Y.-S., Lindzen, R. S., Ho, C., and Kim, J. Space observations of cold-cloud phase change. *Proceedings of the National Academy of Sciences*, 107:11211 – 11216, 2010. URL <https://api.semanticscholar.org/CorpusID:15035574>.
- Curcio, F., Castro, P., Fonseca, A., Castro, R., Franco, R., Ogasawara, E., Stepanenko, V., Porto, F., Ferro, M., and Bezerra, E. Towards a spatiotemporal fusion approach to precipitation nowcasting. In *2025 28th International Conference on Information Fusion (FUSION)*, pp. 1–9, 2025. doi: 10.23919/FUSION65864.2025.11123942.
- Dacre, H. F., Clark, P. A., Martínez-Alvarado, O., Stringer, M., and Lavers, D. A. How do atmospheric rivers form. *Bulletin of the American Meteorological Society*, 96:1243–1255, 2015. URL <https://api.semanticscholar.org/CorpusID:21602930>.
- Davolio, S., Ferretti, R., Flamant, C., and agustin Jansa. Hymex-sop1 the field campaign dedicated to heavy precipitation and flash flooding in the northwestern mediterranean. 2018. URL <https://api.semanticscholar.org/CorpusID:276605697>.
- Durrán, D. R. Can the issuance of hazardous-weather warnings inform the attribution of extreme events to climate change? *Bulletin of the American Meteorological Society*, 101(8):E1452–E1463, 2020.
- Emanuel, K. A. *Atmospheric convection*. Oxford university press, 1994.
- Feingold, G., Koren, I., Wang, H., Xue, H., and Brewer, W. A. Precipitation-generated oscillations in open cellular cloud fields. *Nature*, 466(7308):849–852, 2010.

- Gao, Z., Shi, X., Wang, H., Zhu, Y., Wang, Y. B., Li, M., and Yeung, D.-Y. Earthformer: Exploring space-time transformers for earth system forecasting. *Advances in Neural Information Processing Systems*, 35:25390–25403, 2022a.
- Gao, Z., Tan, C., Wu, L., and Li, S. Z. Simvp: Simpler yet better video prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3170–3180, 2022b.
- Gao, Z., Shi, X., Han, B., Wang, H., Jin, X., Maddix, D. C., Zhu, Y., Li, M., and Wang, B. Prediff: Precipitation nowcasting with latent diffusion models. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.
- Guen, V. L. and Thome, N. Disentangling physical dynamics from unknown factors for unsupervised video prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11474–11484, 2020.
- Guo, Y., Yang, C., Rao, A., Liang, Z., Wang, Y., Qiao, Y., Agrawala, M., Lin, D., and Dai, B. Animatediff: Animate your personalized text-to-image diffusion models without specific tuning. *International Conference on Learning Representations*, 2024.
- Houze Jr, R. A. Mesoscale convective systems. *Reviews of Geophysics*, 42(4), 2004.
- Jordi, A., Georgas, N., Blumberg, A., Yin, L., Chen, Z., Wang, Y., Schulte, J., Ramaswamy, V., Runnels, D., and Saleh, F. A next-generation coastal ocean operational system: Probabilistic flood forecasting at street scale. *Bulletin of the American Meteorological Society*, 100(1): 41–54, 2019.
- Kalnay, E. *Atmospheric modeling, data assimilation and predictability*. Cambridge university press, 2003.
- Kim, W., Jeong, C.-H., and Kim, S. Improvements in deep learning-based precipitation nowcasting using major atmospheric factors with radar rain rate. *Computers & Geosciences*, 184:105529, 2024.
- Kingma, D. P. and Welling, M. Auto-encoding variational bayes. *CoRR*, abs/1312.6114, 2013. URL <https://api.semanticscholar.org/CorpusID:216078090>.
- Lam, R., Sanchez-Gonzalez, A., Willson, M., Wirnsberger, P., Fortunato, M., Alet, F., Ravuri, S. V., Ewalds, T., Eaton-Rosen, Z., Hu, W., Merose, A., Hoyer, S., Holland, G., Vinyals, O., Stott, J., Pritzel, A., Mohamed, S., and Battaglia, P. Learning skillful medium-range global weather forecasting. *Science*, 382:1416 – 1421, 2023. URL <https://api.semanticscholar.org/CorpusID:265156864>.
- Lin, K., Zhang, B., Yu, D., Feng, W., Chen, S., Gao, F., Li, X., and Ye, Y. Alphapre: Amplitude-phase disentanglement model for precipitation nowcasting. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 17841–17850, 2025.
- Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. *ArXiv*, abs/2210.02747, 2022. URL <https://api.semanticscholar.org/CorpusID:252734897>.
- MeteoNet. Meteonet. <https://meteofrance.github.io/meteonet/>, 2021. Accessed: 2026-01-14.
- Nearing, G., Cohen, D., Dube, V., Gauch, M., Gilon, O., Harrigan, S., Hassidim, A., Klotz, D., Kratzert, F., Metzger, A., et al. Global prediction of extreme floods in ungauged watersheds. *Nature*, 627(8004):559–563, 2024.
- NEWS. Texas flooding updates: Death toll reaches 134, search continues for missing. <https://abcnews.go.com/US/live-updates/texas-flooding-live-updates/?id=123729682>, 2025. Accessed: 2025-10-29.
- NOAA. Noaa global historical climatology network hourly (ghcnh). <https://www.ncsl.noaa.gov/products>. Accessed: 2026-01-14.
- Pan, X., Lu, Y., Zhao, K., Huang, H., Wang, M., and Chen, H. Improving nowcasting of convective development by incorporating polarimetric radar variables into a deep-learning model. *Geophysical Research Letters*, 48(21): e2021GL095302, 2021.
- Parker, D. J., Willetts, P., Birch, C., Turner, A. G., Marsham, J. H., Taylor, C. M., Kolusu, S., and Martin, G. M. The interaction of moist convection and mid-level dry air in the advance of the onset of the indian monsoon. *Quarterly Journal of the Royal Meteorological Society*, 142(699): 2256–2272, 2016.
- Peebles, W. S. and Xie, S. Scalable diffusion models with transformers. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4172–4182, 2022. URL <https://api.semanticscholar.org/CorpusID:254854389>.
- Peng, W., Bao, S., Yang, K., Wei, J., Zhu, X., Qiao, Z., Wang, Y., and Li, Q. Radar quantitative precipitation estimation algorithm based on precipitation classification and dynamical zr relationship. *Water*, 14(21):3436, 2022.

- Perez, E., Strub, F., de Vries, H., Dumoulin, V., and Courville, A. C. Film: Visual reasoning with a general conditioning layer. In *AAAI Conference on Artificial Intelligence*, 2017. URL <https://api.semanticscholar.org/CorpusID:19119291>.
- Pressure-level, E. Era5 hourly data on pressure levels from 1940 to present. <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels?tab=download>. Accessed: 2026-01-14.
- Ravuri, S., Lenc, K., Willson, M., Kangin, D., Lam, R., Mirowski, P., Fitzsimons, M., Athanassiadou, M., Kashem, S., Madge, S., et al. Skilful precipitation nowcasting using deep generative models of radar. *Nature*, 597(7878):672–677, 2021.
- Reichstein, M., Benson, V., Blunk, J., Camps-Valls, G., Creutzig, F., Fearnley, C. J., Han, B., Kornhuber, K., Rahaman, N., Schölkopf, B., et al. Early warning of complex climate risk with integrated artificial intelligence. *Nature Communications*, 16(1):2564, 2025.
- Runge, J., Bathiany, S., Bollt, E., Camps-Valls, G., Coumou, D., Deyle, E., Glymour, C., Kretschmer, M., Mahecha, M. D., Muñoz-Marí, J., et al. Inferring causation from time series in earth system sciences. *Nature communications*, 10(1):2553, 2019.
- Schumacher, C., Houze, R. A., and Kraucunas, I. P. The tropical dynamical response to latent heating estimates derived from the trmm precipitation radar. *Journal of the Atmospheric Sciences*, 61:1341–1358, 2004. URL <https://api.semanticscholar.org/CorpusID:55670841>.
- Schumacher, R. S. and Rasmussen, K. L. The formation, character and changing nature of mesoscale convective systems. *Nature Reviews Earth & Environment*, 1(6): 300–314, 2020.
- Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K., and Woo, W.-c. Convolutional lstm network: A machine learning approach for precipitation nowcasting. *Advances in neural information processing systems*, 28, 2015.
- Single-level, E. Era5 hourly data on single levels from 1940 to present. <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview>. Accessed: 2026-01-14.
- Squitieri, B. J., Wade, A. R., and Jirak, I. L. A historical overview on the science of derechos. part 2: Parent storm structure, environmental conditions, and history of numerical forecasts. *Bulletin of the American Meteorological Society*, 2023. URL <https://api.semanticscholar.org/CorpusID:261408777>.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., and Wojna, Z. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2818–2826, 2016.
- Veillette, M., Samsi, S., and Mattioli, C. Sevir: A storm event imagery dataset for deep learning applications in radar and satellite meteorology. *Advances in Neural Information Processing Systems*, 33:22009–22019, 2020.
- Wang, Y., Wu, H., Zhang, J., Gao, Z., Wang, J., Yu, P. S., and Long, M. Predrnn: A recurrent neural network for spatiotemporal predictive learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2):2208–2225, 2022.
- Williamson, C. J., Kupc, A., Axisa, D., Bilsback, K. R., Bui, T., Campuzano-Jost, P., Dollner, M., Froyd, K. D., Hodshire, A. L., Jimenez, J. L., et al. A large source of cloud condensation nuclei from new particle formation in the tropics. *Nature*, 574(7778):399–403, 2019.
- Wu, H., Liang, Y., Xiong, W., Zhou, Z., Huang, W., Wang, S., and Wang, K. Earthfarsser: Versatile spatio-temporal dynamical systems modeling in one model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 15906–15914, 2024a.
- Wu, H., Xu, F., Chen, C., Hua, X.-S., Luo, X., and Wang, H. Pastnet: Introducing physical inductive biases for spatio-temporal video prediction. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 2917–2926, 2024b.
- Yu, D., Li, X., Ye, Y., Zhang, B., Luo, C., Dai, K., Wang, R., and Chen, X. Diffcast: A unified framework via residual diffusion for precipitation nowcasting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 27758–27767, 2024.
- Zhang, Y., Long, M., Chen, K., Xing, L., Jin, R., Jordan, M. I., and Wang, J. Skilful nowcasting of extreme precipitation with nowcastnet. *Nature*, 619(7970):526–532, 2023.

A. StormBench

A.1. Variable in StormBench

Table 3. Multi-source observational data in StormBench.

Region	Source	Variable	Description	Δt
USA	Satellite	VIS	0.6 μm visible satellite channel	5 min
	Satellite	IR069	6.9 μm infrared channel (water vapor)	5 min
	Satellite	IR107	10.7 μm infrared satellite channel	5 min
	Radar	VIL	Vertically integrated liquid from NEXRAD radar	5 min
	Satellite (GLM)	LGHT	Geostationary Lightning Mapper observations	5 min
	Weather Station	Temperature	Near-surface air temperature	1 h
	Weather Station	Pressure	Mean sea-level pressure	1 h
	Weather Station	Wind_s	Wind speed	1 h
	Weather Station	Wind_d	Wind direction	1 h
	Weather Station	Humidity	Relative humidity	1 h
	Weather Station	Precipitation	Accumulated surface precipitation	1 h
FR	Satellite	VIS	0.6 μm visible satellite channel	1 h
	Satellite	IR039	3.9 μm infrared satellite channel	1 h
	Satellite	WV062	6.2 μm infrared channel (water vapor)	1 h
	Satellite	IR108	10.8 μm infrared satellite channel	1 h
	Radar	CR	Radar reflectivity	5 min
	Weather Station	Temperature	Near-surface air temperature	5 min
	Weather Station	Pressure	Mean sea-level pressure	5 min
	Weather Station	Wind_s	Wind speed	5 min
	Weather Station	Wind_d	Wind direction	5 min
	Weather Station	Humidity	Relative humidity	5 min
	Weather Station	Precipitation	Accumulated surface precipitation	5 min

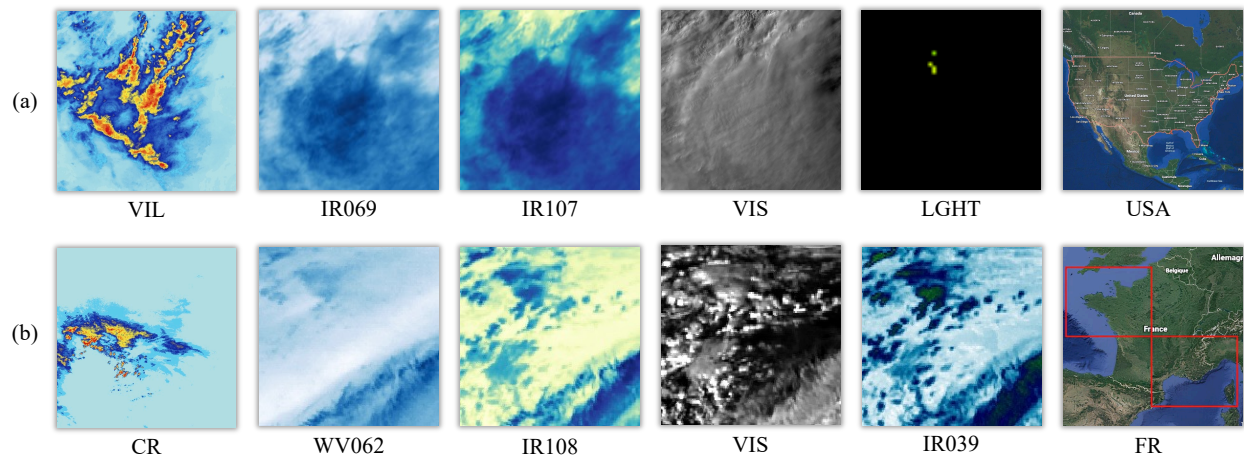


Figure 4. Visualization of observational data across regions: (a) United States (b) France.

To study the relationships among heterogeneous meteorological variables relevant to convective development, we introduce StormBench, a new benchmark that integrates operational multi-source observations with reanalysis fields, covering different climate types regions. Guided by domain experts, StormBench combines satellite, radar, and surface-station measurements with 65 reanalysis variables most closely linked to convective initiation, organization, and evolution, enabling joint analysis of how observational signals and large-scale environmental conditions shape storm dynamics.

Observational Data. StormBench aggregates observational data by region including USA and France, with each region constructed from a multi-source combination of satellite, radar, and surface-station observations, as shown in the Table 3.

- **Surface Observations.** A shared set of near-surface variables is included across regions, comprising air temperature (Temperature), mean sea-level pressure (Pressure), wind speed and direction (Wind.s, Wind.d), relative humidity (Humidity), and accumulated surface precipitation (Precipitation). These variables characterize boundary-layer thermodynamics and surface forcing, which jointly govern the initiation and maintenance of convection through buoyancy generation and moisture supply.
- **Remote Observations.** Higher-level remote-sensing observations contribute information about storm structure, vertical development, and mesoscale organization. Weather radars provide fields such as vertically integrated liquid water content (VIL) or radar reflectivity (CR), describing hydrometeor vertical loading and precipitation intensity—key signatures for identifying convective cores, stratiform regions, and upscale growth. At broader scales, satellite sensors supply visible imagery (VIS) together with thermal infrared and water-vapor channels that capture cloud top morphology, anvil extent, moisture transport pathways, and radiative cooling patterns linked to convective destabilization. Lightning observations (LGHT), when available, offer an additional proxy for convective electrification and deep updraft vigor. The visualisation of remote sensing observation data is shown in Fig. 4.

Due to differences in observing systems, variables are provided at region-dependent temporal resolutions ranging from five minutes to one hour. Collectively, these observational variables provide a physically grounded, multi-modal description of convective evolution spanning surface forcing, vertical structure, and cloud-top processes. This observational data of StormBench is essential for studying real-world storm dynamics and for benchmarking models under heterogeneous regional sensing conditions.

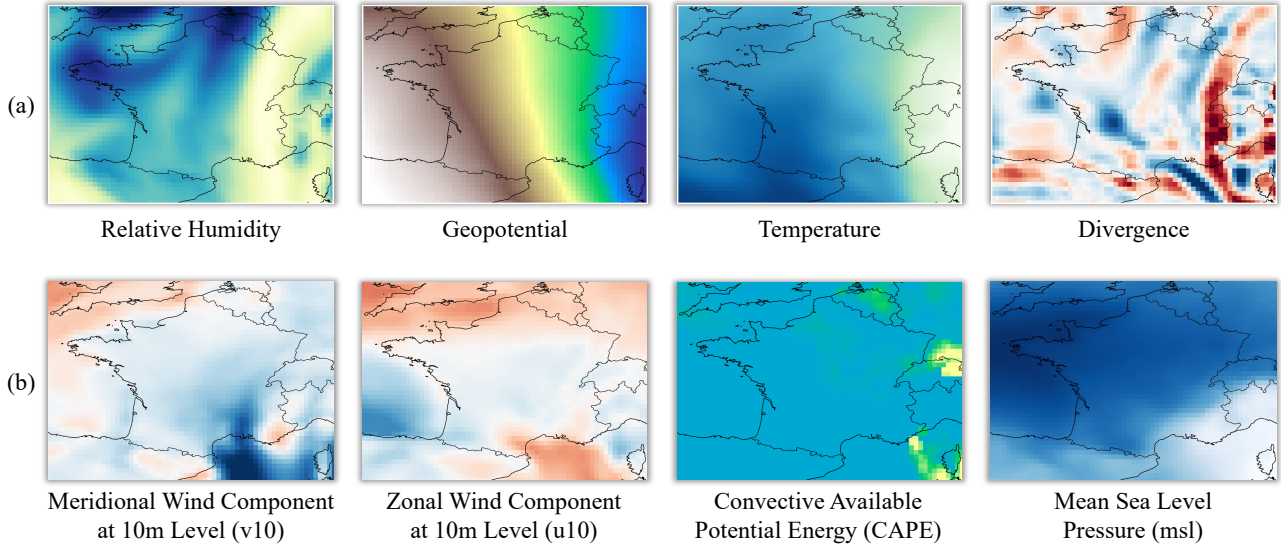


Figure 5. Visualization of partial reanalysis data across (a) Pressure Level (b) Single Level.

Reanalysis Data. Convective initiation is governed by complex three-dimensional atmospheric conditions, including low-level moisture convergence and mesoscale lifting mechanisms such as boundary-layer convergence lines, gust fronts, and orographic forcing. To provide consistent environmental context for observed convective structures, StormBench integrates ERA5 reanalysis from the US and France (Table 4). Provided at hourly cadence ($\Delta t = 1$ h) by ECMWF, these variables capture large-scale and mesoscale thermodynamic and dynamical conditions. As summarized in Table 4, ERA5 fields are organized into two levels, representing pressure-level and near-surface/column-integrated variables, selected under the guidance of meteorologists.

- **ERA5 on Pressure Levels.** The first consists of pressure-level fields on five standard levels (925, 850, 700, 500, and 300 hPa), including temperature, humidity (specific and relative), geopotential height, horizontal divergence, horizontal wind

Table 4. ERA5 reanalysis variables used in StormBench for the USA and France.

Note: Variables shown in *italics* are provided at five pressure levels (300, 500, 700, 850, and 925 hPa).

Source	Variable	Description	Δt
ECMWF	<i>divergence</i>	Horizontal wind divergence (pressure levels)	1 h
	<i>geopotential</i>	Geopotential height (pressure levels)	1 h
	<i>relative_humidity</i>	Relative humidity (pressure levels)	1 h
	<i>specific_humidity</i>	Specific humidity (pressure levels)	1 h
	<i>temperature</i>	Air temperature (pressure levels)	1 h
	<i>u_component_of_wind</i>	Zonal wind component (pressure levels)	1 h
	<i>v_component_of_wind</i>	Meridional wind component (pressure levels)	1 h
	<i>vertical_velocity</i>	Vertical velocity in pressure coordinates	1 h
	10m_u_component_of_wind	Zonal wind component at 10 m above surface	1 h
	10m_v_component_of_wind	Meridional wind component at 10 m above surface	1 h
	100m_u_component_of_wind	Zonal wind component at 100 m above surface	1 h
	100m_v_component_of_wind	Meridional wind component at 100 m above surface	1 h
	2m_temperature	Near-surface air temperature at 2 m	1 h
	2m_dewpoint_temperature	Near-surface dew point temperature at 2 m	1 h
	mean_sea_level_pressure	Pressure reduced to mean sea level	1 h
	instantaneous_10m_wind_gust	Maximum instantaneous wind gust at 10 m	1 h
	surface_latent_heat_flux	Surface latent heat flux	1 h
	surface_net_solar_radiation	Net solar radiation at the surface	1 h
	surface_sensible_heat_flux	Surface sensible heat flux	1 h
	cloud_base_height	Height of cloud base	1 h
	total_column_cloud_liquid_water	Vertically integrated cloud liquid water	1 h
	total_column_water	Total column-integrated water content	1 h
	total_column_water_vapour	Total column-integrated water vapor	1 h
	evaporation	Accumulated surface evaporation	1 h
	convective_precipitation	Convective precipitation	1 h
	large_scale_precipitation	Stratiform (large-scale) precipitation	1 h
	boundary_layer_height	Atmospheric boundary layer height	1 h
	convective_available_potential_energy	Convective available potential energy (CAPE)	1 h
	convective_inhibition	Convective inhibition (CIN)	1 h
	friction_velocity	Surface friction velocity	1 h
	angle_of_sub_gridscale_orography	Orientation angle of sub-grid scale orography	1 h
	anisotropy_of_sub_gridscale_orography	Anisotropy of sub-grid scale orography	1 h
	slope_of_sub_gridscale_orography	Mean slope of sub-grid scale orography	1 h

components (u, v), and vertical velocity. These three-dimensional fields describe synoptic-to-mesoscale lifting, thermal stratification, moisture profiles, and vertical motion—key factors regulating the organization, intensity, and persistence of convection.

- **ERA5 on Single Levels.** The second level consists of near-surface and column-integrated quantities, characterizing boundary-layer conditions (e.g., 2-m temperature and dew point, mean sea-level pressure, 10-m/100-m winds and gusts, sensible/latent heat fluxes, net solar radiation, boundary-layer height), hydrological and cloud-related variables (e.g., total column water/water vapor, cloud-base height, total column cloud liquid water, evaporation, and stratiform/convective precipitation components), and static surface properties (e.g., slope and anisotropy). These variables capture how boundary-layer forcing, moisture transport, and orographic lifting define the environmental pathways through which convection is initiated, sustained, and dissipated. A visualization example of partial reanalysis data attributes is shown in Fig. 5.

Taken together, the ERA5 integration enables joint use of (i) direct storm signatures from operational observations and (ii) physically interpretable environmental controls from reanalysis. This combination provides region-consistent learning signals across heterogeneous national observing systems while preserving the physical links between environmental forcing and storm-scale behavior.

A.2. Spatiotemoral Coverage and Resolution

Event-based Data (US). In the US region, the raw data are sourced from NOAA GHCN_h(NOAA), the "Storm Event Image Set (SEVIR)" dataset(Veillette et al., 2020), and the Climate Data Store(Single-level; Pressure-level). This event-based subset is meticulously curated to organize discrete storm events across the CONUS domain, as shown in Figure 6, spanning multiple Köppen climate types. Each sample is a temporally synchronized sequence containing multi-sensor and multi-attribute reanalysis data, covering a fixed spatial area of $384 \text{ km} \times 384 \text{ km}$ and a 4-hour time window, with a 5-minute sampling interval (49 frames per event). The multivariate three-dimensional atmospheric fields from ERA5 (single level and pressure level) are provided on a regular grid with a resolution of 0.25° , while sparse raw observational data from surface meteorological stations are available within the event coverage area. In addition to storm-event sampling linked to NOAA Storm Events metadata, dataset also includes probability-weighted random sampling across 2017 to 2019 to maintain coverage of diverse precipitation regimes.

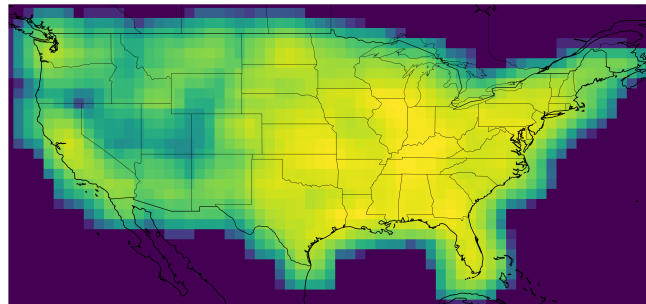


Figure 6. Spatial coverage of event-based data: the brighter the colour on the heatmap, the more events it represents.

Long-term Data (France). In France, the original observational data comes from the Météo-France(MeteoNet, 2021), which is organized as continuous, long-horizon time series rather than storm-centered events. The reanalysis data is also sourced from the Climate Data Store(Single-level; Pressure-level). The observational data cover two fixed geographic regions in France (northwestern region: $46.25^\circ\text{--}51.896^\circ\text{N}$, $-5.842^\circ\text{--}2.0^\circ\text{E}$; southeastern region: $41.10^\circ\text{--}46.25^\circ\text{N}$, $2.0^\circ\text{--}9.842^\circ\text{E}$), each spanning approximately $550 \text{ km} \times 550 \text{ km}$, as shown in Figure 7. The records span from 2016 to 2018 and encompass both temperate oceanic and Mediterranean climate types. Radar rainfall mosaics are provided at 5-minute cadence with a 0.01° grid, offering high-frequency monitoring of precipitation evolution suitable for convective-scale nowcasting and verification. The primary satellite channels used in this work (e.g., VIS, IR108, WV062) are distributed at 0.03° spatial resolution with an hourly cadence (with occasional missing timestamps inherited from upstream archives). The multivariate atmospheric fields from ERA5 (single level and pressure level) are provided on a regular 0.25° grid, aligned with the same spatial and temporal coverage. Together, these design choices yield a multi-year observation set.



Figure 7. Spatial coverage of long-term data.

A.3. Data Construction

Event-based Data Construction. The US subset of our dataset is derived from the SEVIR archive, which provides event-centric collections of satellite and radar tracks. We retain only events where all required remote-sensing modalities (VIS, VIL, IR069, IR107) are simultaneously available, ensuring modality completeness for multi-sensor learning. Surface-station observations from GHCN are ingested for stations located within each event bounding box and aligned to the SEVIR 5-minute cadence via nearest timestamp matching. Point measurements are densified onto the event grid using inverse distance weighting (IDW), yielding gridded near-surface fields co-registered with SEVIR. ERA5 reanalysis variables are extracted over the event spatial extent and interpolated onto the SEVIR grid. To handle native null values, we construct a validity mask and temporarily replace missing entries with event-wise means prior to cubic-spline interpolation; the mask is reapplied afterward to restore missingness semantics. ERA5 fields are then temporally matched to the 5-minute cadence by nearest-neighbor replication for the baseline setting.

Long-term Data Construction. The France subset of our dataset is sourced from the MeteoNet observation system, which provides continuous radar, satellite and station observations from two regions (NW and SE) between 2016 and 2018. The radar reflectivity blocks are concatenated and time-sorted into a strictly increasing sequence, with the native missing timestamps preserved as metadata. The satellite channels (VIS06, WV062, IR108 and IR039) are reindexed to create a complete hourly timeline, which is then filled using forward and backward propagation to ensure a shared time axis. Surface stations are filtered based on data availability and re-indexed to a five-minute grid using forward fill and backward fill for leading gaps, in order to obtain consistent near-surface variables. ERA5 pressure-level and single-level fields are concatenated across pressure levels and stacked along channels to maintain the native spatial grid of each zone.

Unified Sample Construction. To enable standardized benchmarking across the heterogeneous SEVIR and MeteoNet archives, we implement a unified data loading interface that harmonizes the processed modalities into synchronized input-target pairs. The consolidated dataset is partitioned into training, validation, and testing subsets using stratified sampling to ensure distributional consistency. For each sample, the continuous multi-modal timeline is segmented into a fixed sequence: the initial one-hour frames serve as the historical input context, while the subsequent frames constitute the prediction target (see Figure 8). Crucially, despite the differing construction pipelines, all modalities including the IDW-densified surface fields and the upsampled reanalysis data, are aligned to the 5-minute cadence and spatially co-registered before being batched for model training.

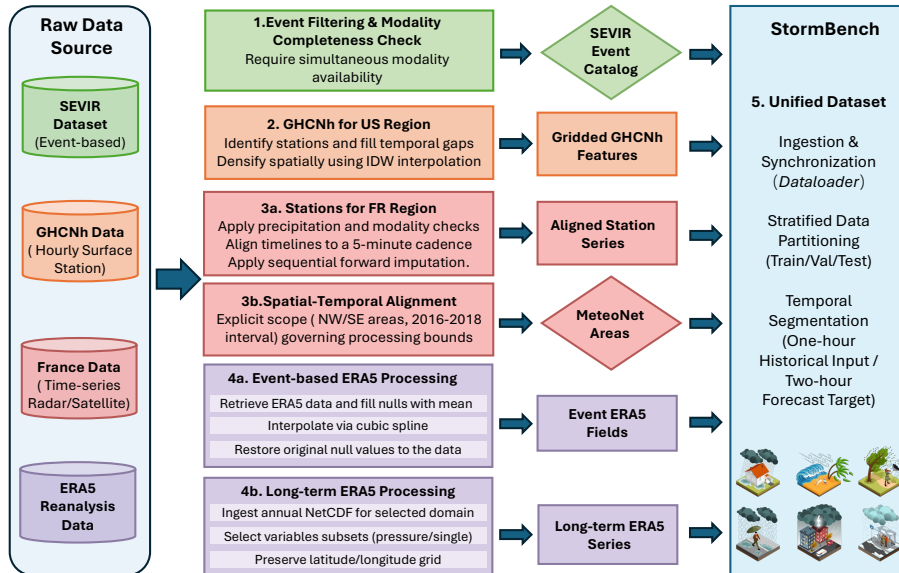


Figure 8. The Pipeline of Data Construction Process.

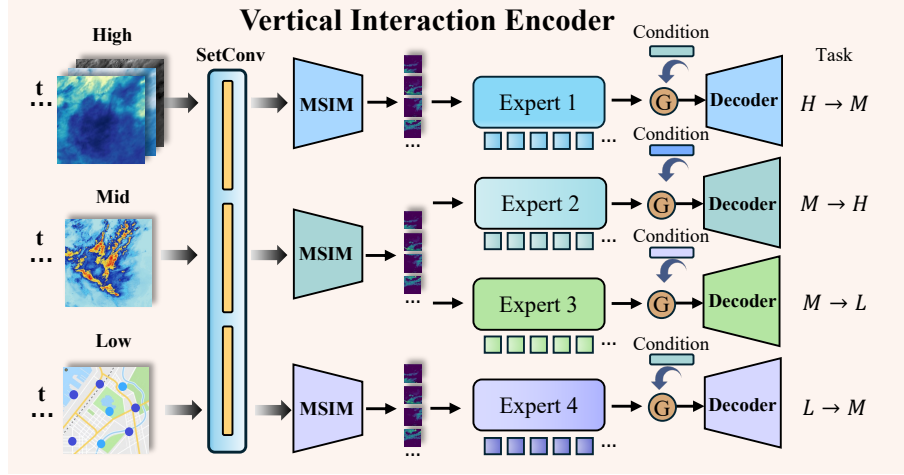


Figure 9. The Structure of Vertical Interaction Encoder.

B. Mixture of Experts Details

Guided by cloud-convective feedback mechanisms (Bony et al., 2015), we introduce a *Vertical Interaction Encoder* (VIE) to capture the directional coupling between atmospheric layers, as shown in Figure 9. Specifically, the VIE models the cross-layer pathways during convective evolution through a set of structured transition operators $\mathcal{E}_{A \rightarrow B}$, representing the influence from layer A to layer B . We conceptually map the heterogeneous input modalities to distinct tropospheric layers:

- **Boundary Layer:** Represented by station data $\mathbf{X}_{sta}$ (e.g., surface temperature, pressure), which governs convective initiation;
- **Mid Convection Layer:** Represented by radar reflectivity $\mathbf{X}_{rad}$, capturing the evolving precipitating cores and hydrometeor dynamics;
- **Upper Troposphere:** Represented by satellite data $\mathbf{X}_{sat}$ (e.g., IR channels), characterizing cloud-top height and anvil spreading.

We define four primary interaction pathways corresponding to distinct meteorological processes:

- **Mid-to-High ($\mathcal{E}_{M \rightarrow L}$):** Captures how mid-level moisture convergence and latent heat release enhance updraft buoyancy, accelerating the vertical transport of air masses to the upper troposphere.
- **High-to-Mid ($\mathcal{E}_{H \rightarrow M}$):** Models top-down feedbacks, such as anvil spreading and radiative cooling at cloud tops, which can modulate mid-level upward motion and trigger secondary convection.
- **Mid-to-Low ($\mathcal{E}_{M \rightarrow L}$):** Represents the disruption of boundary-layer thermodynamics by convective downdrafts and mid-level inflows, reshaping near-surface convergence structures.
- **Low-to-Mid ($\mathcal{E}_{L \rightarrow M}$):** Encodes surface-driven initiation where boundary-layer heating and moistening weaken convective inhibition, triggering new convective cells.

Formally, each interaction is implemented as a directional feature mapping: $\mathbf{H}_B = \text{VIE}(\mathbf{H}_B, \mathbf{H}_A)$, where the hidden state of the target layer is conditioned on the source layer’s representation. This hierarchical design allows the model to explicitly reason about the vertical structure of storm systems, moving beyond simple 2D spatial correlations. To encode these directional pathways, we design an expert-driven MoE component (Section 3.3.2), as shown in Fig. 9.

Training of Expert. The MoE framework is trained independently of StormInsight. Each expert is trained with a reconstruction loss:

$$\mathcal{L}_{i \rightarrow j} = \frac{1}{K} \sum_{t=T}^{T+K} \left\| \hat{\mathbf{X}}_t^j - \mathbf{X}_t^j \right\|_2^2, t \in [T, T+K] \quad (23)$$

When training `StormInsight`, the output embeddings of MoE experts and GoT will be injected into DiT through AdaLN as conditions. The potential trajectories will be influenced by the vertical inter-layer interactions, which is consistent with the multi-scale structure of convective evolution.

C. Training and Inference of Flow Matching

Algorithm 1 Flow Training Process of `StormInsight`

Input: Dataset $\mathcal{D}$, Pre-trained VAE Encoder $\mathcal{E}$, Conditioning Encoder $\mathcal{C}$, Flow Model v_θ , std σ
 Initialize model parameters θ
while not converged **do**
 Sample $(X_{\text{past}}, X_{\text{future}}, O_{\text{cond}}) \sim \mathcal{D}$
 Encode past and future: $\mathbf{z}_{\text{sys}} \leftarrow \mathcal{E}(X_{\text{past}})$, $\mathbf{z}_{\text{future}} \leftarrow \mathcal{E}(X_{\text{future}})$
 Extract conditioning: $\mathbf{Z}_{\text{cond}} = \{\mathbf{z}_{\text{sys}}, \mathbf{z}_{\text{env}}, \mathbf{z}_{\text{expert}}\} \leftarrow \mathcal{C}(O_{\text{cond}})$
 Sample prior and interpolation variables: $\mathbf{z}_0 \sim \mathcal{N}(0, I)$, $t \sim \mathcal{U}[0, 1]$, $\epsilon \sim \mathcal{N}(0, I)$
 Construct interpolated latent: $\mathbf{z}_t \leftarrow (1 - t)\mathbf{z}_0 + t\mathbf{z}_{\text{future}} + \sigma\epsilon \{\text{Optimal transport path} + \text{optional noise}\}$
 Compute target velocity: $\mathbf{u}_t \leftarrow \mathbf{z}_{\text{future}} - \mathbf{z}_0$
 Predict velocity: $\hat{\mathbf{u}}_t \leftarrow v_\theta(\mathbf{z}_t, t, \mathbf{Z}_{\text{cond}})$
 Compute loss: $\mathcal{L}_{\text{FM}} \leftarrow \|\hat{\mathbf{u}}_t - \mathbf{u}_t\|_2^2$
 Update parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{FM}}$
end while

Algorithm 2 Flow Conditional Latent Sampling of `StormInsight`

Input: Input sequence X_{past} , Trained Flow model v_θ , VAE Encoder/Decoder $\mathcal{E}/\mathcal{D}$, Number of ODE steps S
 Encode past: $\mathbf{z}_{\text{sys}} \leftarrow \mathcal{E}(X_{\text{past}})$
 Compute physical conditions: $\mathbf{Z}_{\text{cond}} = \{\mathbf{z}_{\text{sys}}, \mathbf{z}_{\text{env}}, \mathbf{z}_{\text{expert}}\}$
 Sample initial latent: $\mathbf{z}_0 \sim \mathcal{N}(0, I)$
 Set step size: $\Delta t \leftarrow 1/S$
 Initialize current state: $\mathbf{z}_{\text{curr}} \leftarrow \mathbf{z}_0$
for $i = 0$ **to** $S - 1$ **do**
 Current time: $t \leftarrow i \cdot \Delta t$
 Predict vector field: $\mathbf{v} \leftarrow v_\theta(\mathbf{z}_{\text{curr}}, t, \mathbf{Z}_{\text{cond}})$
 Euler update: $\mathbf{z}_{\text{curr}} \leftarrow \mathbf{z}_{\text{curr}} + \mathbf{v} \cdot \Delta t$
end for
 Final latent: $\mathbf{z}_1 \leftarrow \mathbf{z}_{\text{curr}}$
 Decode to pixel space: $\hat{Y}_{\text{future}} \leftarrow \mathcal{D}(\mathbf{z}_1)$
Output: Predicted future radar $\hat{Y}_{\text{future}}$

D. Experiment Setting Details

D.1. Variables Setting

- **Reanalysis Data.** To ensure a fair comparison across baselines with heterogeneous input requirements, we adapt the reanalysis data selection process across regions. Some baseline models are not designed for multi-modal inputs and cannot accommodate full pressure-level atmospheric fields under GPU memory constraints. To resolve this mismatch, we composite the pressure-level fields and retain only the most relevant single-level variables (see Table 5).
- **Satellite.** Satellite provide radiative measurements that describe cloud-top microphysics and cloud organization.
USA: VIS (0.6 μm), IR069 (6.9 μm), IR107 (10.7 μm).
FR: VIS (0.6 μm), IR039 (3.9 μm), WV062 (6.2 μm), IR108 (10.8 μm).
- **Radar.** Radar captures hydrometeors within precipitating columns and maps convective intensity with spatial fidelity.
USA: Vertically Integrated Liquid (VIL).
FR: Radar Reflectivity (CR).

Table 5. Selected reanalysis variables used for model comparison.

Category	Variables
Pressure-level (composite)	divergence geopotential relative humidity specific humidity temperature u-component of wind v-component of wind vertical velocity
Single-level (expert-select)	10m u-component of wind 10m v-component of wind mean sea level pressure boundary layer height convective available potential energy (CAPE) total column water vapour

- **Weather Stations.** Surface stations provide near-boundary thermodynamic conditions and storm footprints.
USA: Temperature, Pressure, Wind speed, Wind direction, Humidity, Precipitation.
FR: Temperature, Pressure, Wind speed, Wind direction, Humidity, Precipitation.

D.2. Baseline Details

We compare *StormInsight* with eleven competitive baselines, including deterministic and probabilistic baselines. All baselines were retrained under Multi-Input Single-Output (MISO) setting by adjusting the channels and maintaining a comparable computational load to ensure fairness. *Prediff* is not included in the main multi-source comparison because its architectural design does not permit straightforward fusion of non-radar inputs via channel expansion:

- **ConvLSTM** (Shi et al., 2015) models radar evolution as a seq-to-seq video prediction task to capture local spatiotemporal dependencies.
- **PredRNN-V2** (Wang et al., 2022) extends ConvLSTM with spatiotemporal memory cells to better model long-term dependencies.
- **PhyDNet** (Guen & Thome, 2020) implements a physics-inspired architecture that disentangles motion and content via dedicated encoders.
- **SimVP** (Gao et al., 2022b) adopts a minimalist two-stage architecture that separates spatial encoding and temporal forecasting.
- **Earthformer** (Gao et al., 2022a) employs cuboid self-attention to efficiently capture short-range spatiotemporal dependencies.
- **DGMR** (Ravuri et al., 2021) leverages a diffusion-like generative hierarchy combined with spatial discriminators to produce high-resolution and distributionally coherent precipitation nowcasts (Appendix E.3).
- **NowcastNet** (Zhang et al., 2023) integrates dual-domain generators with physical alignment constraints to improve precipitation realism, intensity calibration, and storm initiation skill.
- **PreDiff** (Gao et al., 2023) formulates probabilistic precipitation nowcasting as a conditional latent diffusion process (Appendix E.3).
- **PastNet** (Wu et al., 2024b) employs a multi-path encoder infused with physics-informed inductive biases to strike a balance between prediction accuracy and computational cost.
- **Earthfarseer** (Wu et al., 2024a) advances Earthformer with causality-aware modules and hierarchical fusion for long-range modeling.

- **AlphaPre** (Lin et al., 2025) improves spatial detail preservation by disentangling amplitude and phase components in the frequency domain.

D.3. Evaluation Metrics: Definitions and Formulations

To evaluate both the quantitative accuracy and physical fidelity of the predicted radar fields $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times W}$ against the ground truth $\mathbf{Y}$, we employ the following metrics:

D.3.1. DISTRIBUTIONAL AND NUMERICAL ACCURACY

Mean Absolute Error (MAE) quantify distributional bias; lower values indicate higher numerical accuracy. Measures the average pixel-wise numerical discrepancy:

$$\text{MAE} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W |\hat{y}_{i,j} - y_{i,j}|. \quad (24)$$

D.3.2. CATEGORICAL SKILL AND STORM DETECTION

Given a reflectivity threshold τ (e.g., 30 dBZ), we define the number of hits (H), false alarms (F), and misses (M).

Critical Success Index (CSI) measures the ratio of correctly predicted convective events to the total number of predicted or observed events; higher values indicate more reliable localization of convective structures:

$$\text{CSI} = \frac{H}{H + F + M}. \quad (25)$$

Heidke Skill Score (HSS) assesses the improvement of the forecast over a random chance prediction:

$$\text{HSS} = \frac{2(H \cdot TN - F \cdot M)}{(H + M)(M + TN) + (H + F)(F + TN)}, \quad (26)$$

where TN denotes true negatives.

Thresholds of different area:

USA: 16, 74, 133, 160, 181, 219.

FR: 19, 28, 35, 40, 47.

D.3.3. SPATIAL AND PERCEPTUAL FIDELITY

Peak Signal-to-Noise Ratio (PSNR) evaluates the maximum potential power of the signal relative to the distorting noise; higher values reflect clearer predictions that better respect physical realism:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \quad (27)$$

where MAX_I is the maximum possible pixel value of the radar imagery.

Structural Similarity Index (SSIM) measures the degradation of structural information by comparing luminance (l), contrast (c), and structure (s) between two images:

$$\text{SSIM}(\mathbf{Y}, \hat{\mathbf{Y}}) = [l(\mathbf{Y}, \hat{\mathbf{Y}})]^\alpha \cdot [c(\mathbf{Y}, \hat{\mathbf{Y}})]^\beta \cdot [s(\mathbf{Y}, \hat{\mathbf{Y}})]^\gamma, \quad (28)$$

where $\alpha, \beta, \gamma > 0$ are weights. SSIM is particularly effective at capturing the morphological realism of storm cells.

D.3.4. EVENT-ORIENTED METRICS

To assess the detection of rainfall events (e.g., 0.1 mm/h), we use Probability of Detection (POD) (Durrant, 2020) and False Alarm Ratio (FAR) (Jordi et al., 2019), defined by hits H , missed M , and false alarms F :

$$\text{POD} = \frac{H}{H + M}, \quad \text{FAR} = \frac{F}{H + F}. \quad (29)$$

POD measures the proportion of actual rain events correctly identified, while FAR quantifies the fraction of predicted events that did not occur.

E. Extra Experiment Results

E.1. Statistical Significance Analysis

To ensure empirical rigor and reproducibility, all experiments are run using five independent random seeds. We compare N against the state-of-the-art baseline using the Wilcoxon signed-rank test, a non-parametric method that avoids assumptions about normality. The observed improvements are consistent across seeds, and are statistically significant ($p < 0.05$), which confirms that the reported gains reflect the robust and reproducible performance of our framework, rather than stochastic variation. This combination of multi-seed evaluation and non-parametric testing shows that advancements made by N are statistically sound across different areas.

E.2. Real-Time Inference Efficiency

To evaluate the computational efficiency of StormInsight, we assess their memory footprint and inference latency on a single NVIDIA H100 GPU using the StormBench, as shown in Table 6. Despite incorporating multiple experts, StormInsight achieves a forward inference latency of approximately 380 ms per 25 frames, falling well within practical operational constraints and demonstrating its efficiency and suitability for real-world deployment.

Table 6. Inference speed and memory.

Resolution	Latency (ms)	Peak Memory (MB)	Dynamic Memory (MB)
128×128	372.52 ± 4.10	2121.80	1627.40

E.3. Performance of the baseline relying solely on radar observations

The baseline models used in our study were originally designed for only radar input. To ensure a fair comparison under this setting, we evaluated all baselines using radar-only inputs on our benchmark through the StormBench, and additionally incorporated probabilistic baselines Prediff(Gao et al., 2023) and DGMR(Ravuri et al., 2021). Compared with the baselines, StormInsight under the Multi-Input Single-Output (MISO) setting still achieved excellent performance.

Table 7. Performance comparison across different region. (Baselines only with radar input.)

Model	USA								France							
	MAE↓	mCSI↑	mHSS↑	CSI ₁₆ ↑	CSI ₇₄ ↑	CSI ₂₁₉ ↑	PSNR↑	SSIM↑	MAE↓	mCSI↑	mHSS↑	CSI ₁₉ ↑	CSI ₂₈ ↑	CSI ₄₇ ↑	PSNR↑	SSIM↑
ConvLSTM	15.897	0.326	0.413	0.625	0.522	0.175	23.223	0.484	1.330	0.172	0.260	0.384	0.262	0.030	41.364	0.891
PredRNN-V2	16.570	0.282	0.343	0.606	0.521	0.095	23.176	0.484	1.416	0.157	0.239	0.366	0.235	0.031	41.231	0.883
PhyDNet	17.431	0.322	0.406	0.589	0.517	0.236	22.668	0.399	1.493	0.125	0.190	0.332	0.181	0.023	41.117	0.877
SimVP	14.321	0.387	0.495	0.653	0.561	0.291	23.896	0.543	1.254	0.183	0.270	0.399	0.269	0.051	41.780	0.898
Earthformer	16.717	0.332	0.416	0.617	0.545	0.204	22.884	0.398	1.345	0.118	0.182	0.328	0.178	0.022	41.252	0.888
DGMR	18.452	0.199	0.265	0.447	0.318	0.119	21.652	0.459	1.400	0.098	0.155	0.254	0.148	0.003	39.918	0.861
NowcastNet	17.862	0.291	0.375	0.578	0.477	0.143	21.652	0.459	2.006	0.144	0.228	0.305	0.208	0.148	39.019	0.815
PredDiff	17.261	0.373	0.472	0.613	0.594	0.119	23.672	0.571	2.174	0.108	0.176	0.228	0.163	0.003	34.667	0.815
PastNet	14.640	0.353	0.450	0.637	0.559	0.208	23.530	0.531	1.434	0.156	0.239	0.370	0.232	0.033	41.055	0.877
Earthfarseer	14.621	0.347	0.445	0.647	0.553	0.189	23.396	0.517	1.498	0.134	0.207	0.323	0.179	0.040	40.702	0.871
AlphaPre	15.469	0.351	0.439	0.631	0.545	0.241	23.780	0.532	1.456	0.130	0.196	0.288	0.165	0.048	41.384	0.876
Ours	12.608	0.540	0.632	0.673	0.552	0.361	23.651	0.573	0.980	0.258	0.380	0.448	0.354	0.087	30.089	0.811

E.4. More Visualization Results

We provide additional visualizations of convective development and dissipation processes, where StormInsight consistently demonstrates superior performance. In contrast, existing baselines exhibit significant degradation, primarily due to architectural limitations in integrating multi-modal attributes. Notably, Figure 10 illustrates a challenging case involving a cloud precursor initially outside the radar’s range that abruptly enters the sensing domain; all models, including our own, show limited responsiveness to such stochastic boundary entries. Additionally, benefiting from comprehensive meteorological context, StormInsight demonstrates an enhanced capability to capture sudden convective intensification, which remains a significant challenge for conventional methods.

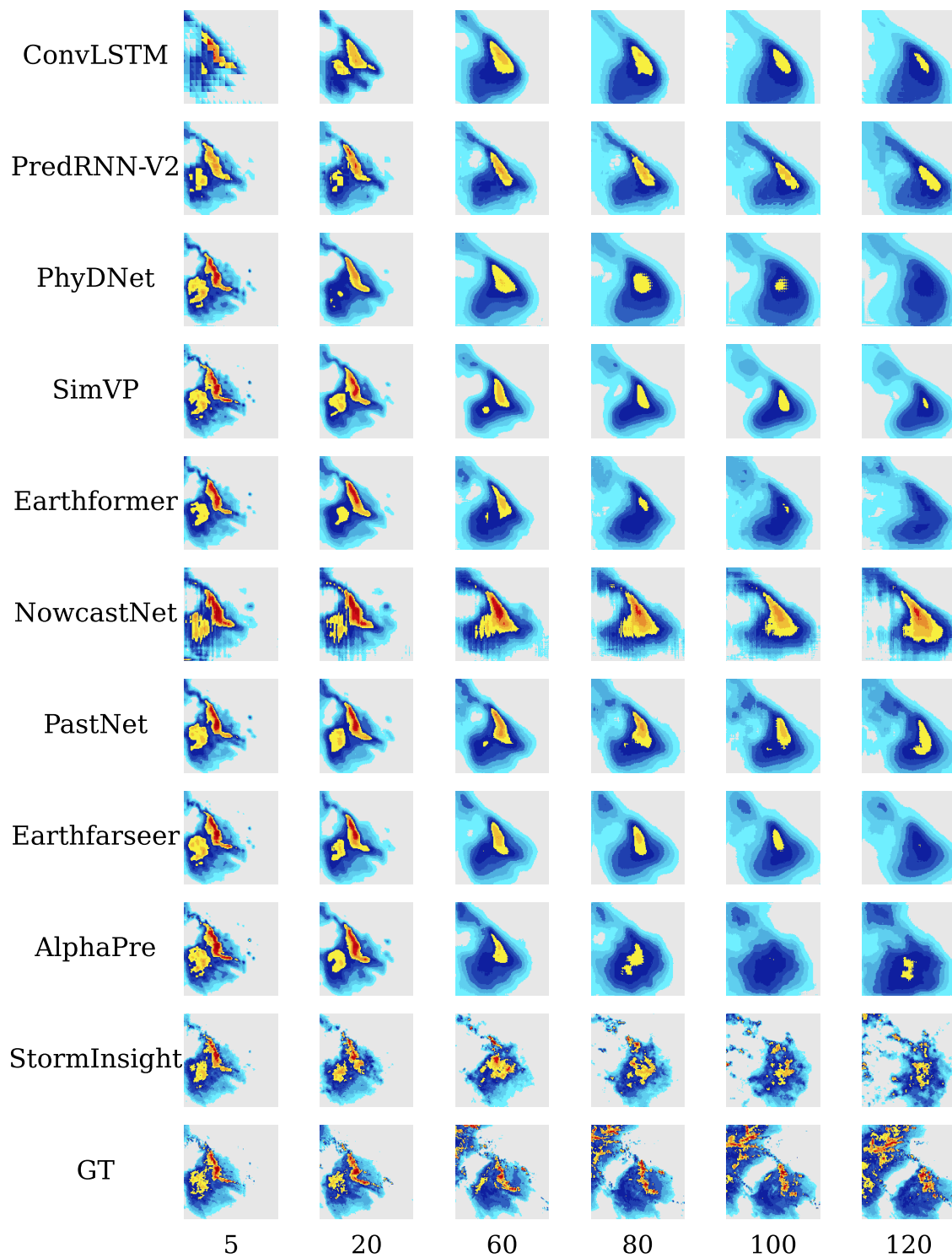


Figure 10. The Convective Development Case.

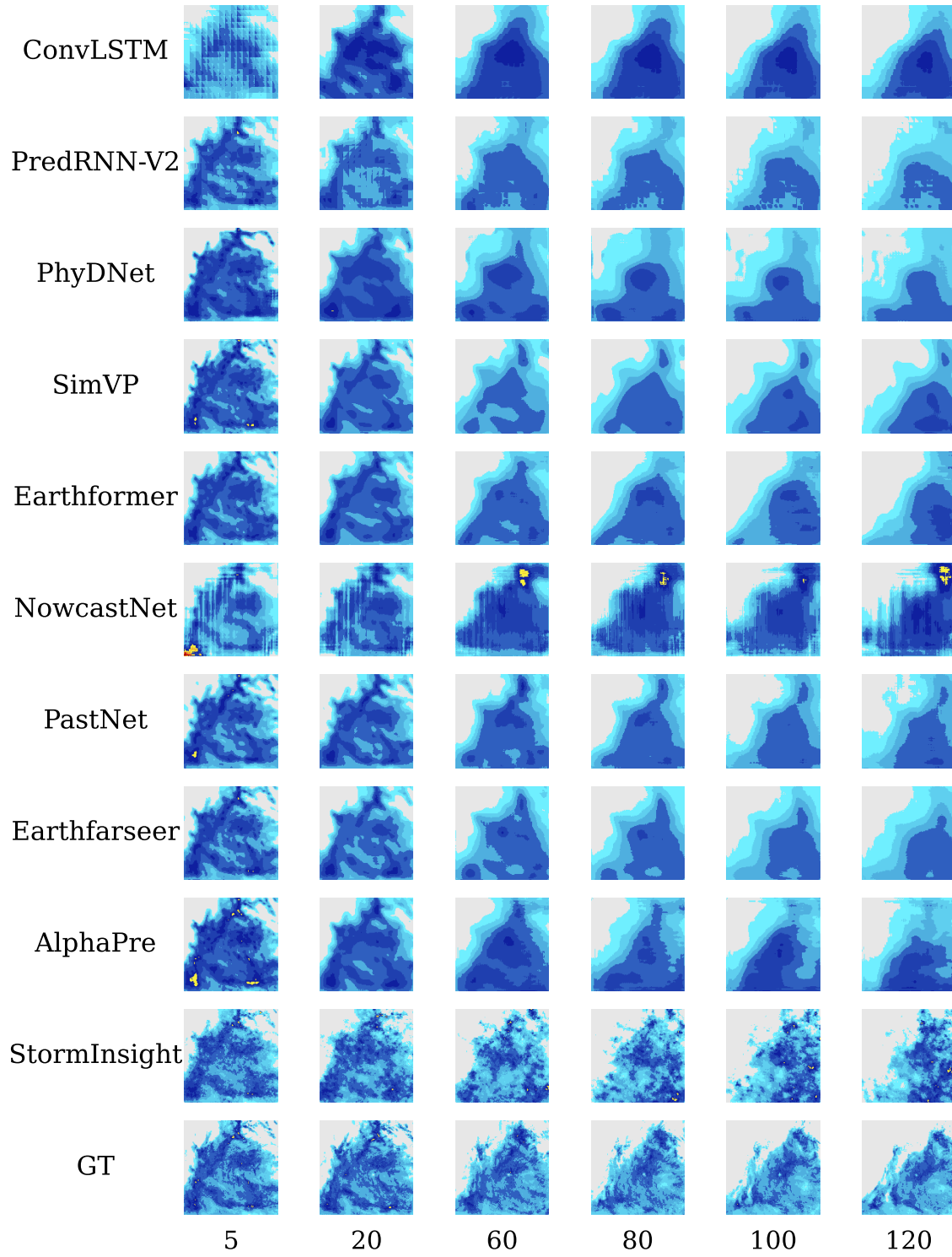


Figure 11. The Convective Dissipation Case.

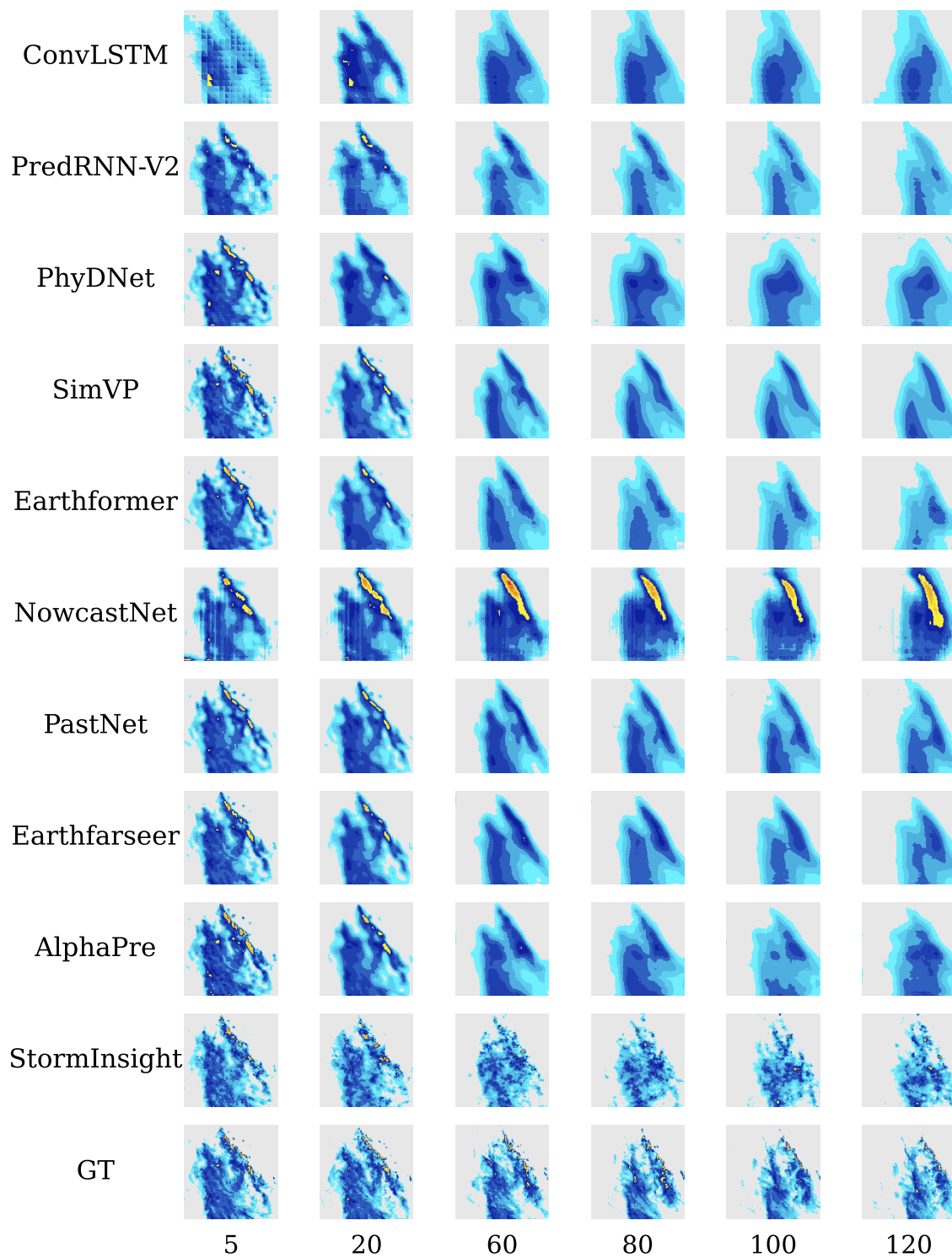


Figure 12. The Convective Dissipation Case.

E.5. Performance of Precipitation Estimation

To assess the downstream utility of *StormInsight*, we conduct preliminary rainfall estimation experiments using the extrapolated radar fields as input. After applying a conventional reflectivity–rainfall conversion Z–R(Peng et al., 2022), the predicted rainfall closely aligns with ground measurements (POD: 0.763, FAR: 0.018), indicating strong warning capability. These results suggest that extrapolated radar fields can effectively support downstream tasks such as rainfall estimation, providing actionable information for risk mitigation, traffic management, and emergency response.

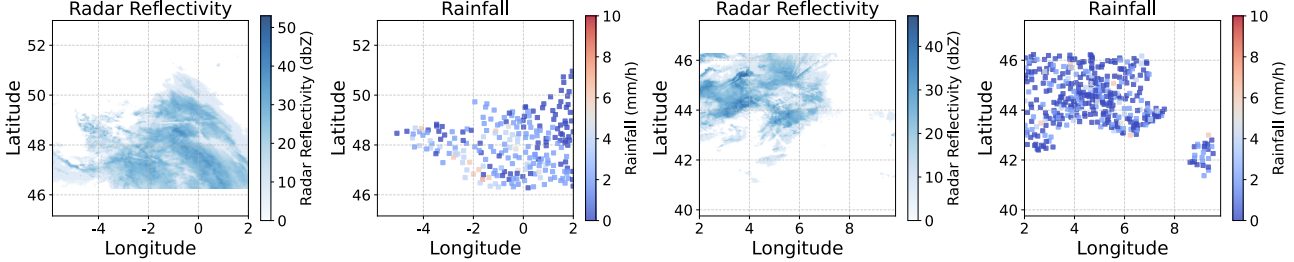


Figure 13. Visualization of Reconstructed Rainfall Field.

F. Discussion on Potential Data Leakage of ERA5 Data

A potential concern when utilizing ERA5 reanalysis data is the indirect information leakage stemming from four-dimensional variational (4D-Var) data assimilation. We address this concern from three perspectives: task formulation, established precedent, and temporal alignment settings.

Indirect Impact of Data Assimilation. Due to 4D-Var assimilation, potential data leakage commonly exists in tasks predicting reanalysis fields. However, unlike global weather models (e.g., Pangu-Weather) that directly predict reanalysis fields as targets, our task focuses on radar echo extrapolation strictly from observations. In our framework, ERA5 is utilized solely as a large-scale conditioning field to provide thermodynamic and dynamic context, rather than serving as the prediction target. Consequently, any potential information leakage from the reanalysis variables is indirect and significantly weaker than in reanalysis-to-reanalysis forecasting. Furthermore, existing literature indicates that discrepancies between reanalysis fields and actual operational initial states typically lead to only marginal performance degradation.

To validate this robust real-world performance, we conducted a pilot study in collaboration with a local meteorological administration during the 2025 rainy season. We replaced the ERA5 conditioning fields with **CMA real-time operational analysis products**. Evaluated using the event-based early warning hit rate for heavy rainfall, our operational model achieved an accuracy of 97.18%, demonstrating strong generalization without relying on future-assimilated reanalysis data.

Established Precedent. Combining local observational data with ERA5 reanalysis is a widely validated practice in meteorological deep learning. Prior studies have successfully employed reanalysis data as a large-scale background field to guide local precipitation evolution and convective initiation (Kim et al., 2024; Curcio et al., 2025).

Temporal Alignment. To strictly prevent access to future information during inference, we enforce rigorous temporal alignment. For forecasting 5-minute radar frames from $t = 0$ to $t = 1$ h, the model exclusively accesses the single ERA5 frame at $t = 0$ via nearest-neighbor replication. We explicitly prohibit temporal interpolation or the inclusion of the ERA5 frame at $t = 1$ h, thereby ensuring zero temporal leakage within the active prediction window.

Ultimately, fully operational real-world deployment relies on non-assimilated operational analysis and forecast fields (e.g., GFS or IFS). The robust results from our real-time pilot study outline a highly feasible and effective path toward such fully operational deployment.